Stones, Obstruction & Trauma Rapid Revision

Campbell-Walsh-Wein · Chapters 54, 62, 93–100, 125

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 5h 45min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 98* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
93	Management of Upper Urinary Tract Obstruction	39 min
94	Upper Urinary Tract Trauma	35 min
95	Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis	57 min
96	Acute Care of Urinary Lithiasis	13 min
97	Surgical Management of Ureteral Calculi	29 min
98	Surgical Management of Upper Urinary Tract Calculi	34 min
99	Prevention of Urinary Stone Disease	21 min
100	Lower Urinary Tract Calculi	13 min
54	Management of Paediatric Kidney Stone Disease	25 min
125	Genital and Lower Urinary Tract Trauma	32 min
62	Paediatric Genitourinary Trauma	37 min
★	Putting it together: stones, obstruction and trauma	10 min
	Total (24237 words, at 70 words/min)	**5h 45min**

93 Management of Upper Urinary Tract Obstruction

Reading time ≈ 39 min (slow pace)

1 · URETEROPELVIC JUNCTION OBSTRUCTION (UPJO): PRESENTATION

- Chapter follows anatomical level: ureteropelvic junction (UPJ) → retrocaval ureter → ureteral stricture → ureteroenteric anastomosis → retroperitoneal fibrosis (RPF). Trauma → *Ch 94, Ch 125*; stones → *Ch 95–100*.
- UPJO is usually congenital but presents at any age. **Neonates**: mostly prenatal hydronephrosis on ultrasonography (historically a flank mass); azotaemia if bilateral or solitary kidney. **Older children/adults**: intermittent flank pain with vomiting, haematuria after minor trauma, urinary tract infection (UTI), rarely hypertension.

2 · DIAGNOSIS

> **KEY POINT** **Dilation is not obstruction.** Anatomical imaging (ultrasonography, computed tomography [CT], retrograde) shows *where*; functional tests (renography, Whitaker) show *whether it matters*. Combine one of each to plan therapy.

Modality	Key point
Ultrasonography (first in neonates)	Large medial pelvis, rounded calyces joined by dilated infundibula → distinguishes from multicystic kidney; ureter not seen
Non-contrast CT (acute flank pain)	Hydronephrosis to UPJ without stone; excludes radiolucent stone, tumour
Contrast CT	Shows **crossing vessels**; can substitute for renography
Diuretic renography	Differential function, drainage; best follow-up study
Retrograde pyelography	Usually **at the time of repair** (avoids introducing infection)
Nephrostomy + antegrade study	Retrograde fails/hazardous; tests recoverability; permits Whitaker

- **Renography**: hydrated → furosemide **0.3–0.5 mg/kg** at **20 min** (collecting system fills first). Technetium-99m mercaptoacetyltriglycine (MAG3) preferred over diethylenetriaminepentaacetic acid (DTPA) for image quality and dosimetry. **Obstructed: drainage half-time (T½) >40 min**; after successful pyeloplasty drainage occurs even before furosemide.
- P40 (% of maximal tracer left at 40 min) raises sensitivity. Renography also excludes coexisting UPJO in high-grade reflux and is the most accurate study after intervention.

ALGORITHM

Whitaker test: nephrostomy → bladder catheter (stops transmission of bladder pressure) → perfuse pelvis at **10 mL/min** → non-obstructed ≤12–15 cm H_2O; obstructed >**15–20 cm H_2O**. Invasive → only when other tests non-diagnostic. Errors: pelvic compliance, patient position

3 · INDICATIONS AND CHOICE OF INTERVENTION

- **Indications**: symptoms; impaired or progressively falling function; stones or infection; rarely hypertension.
- **Neonates**: selective observation supported — operate for progressive hydronephrosis or a **10%** fall in differential function. **Asymptomatic adults**, indeterminate → serial renography.
- **Procedure of choice**: dismembered pyeloplasty, now robotic at most major centres; open = minimally invasive; endopyelotomy fails more.
- Pre-repair drainage (stent or nephrostomy) only for infection, azotaemia (solitary kidney/bilateral) or unrelenting pain.

Endopyelotomy subgroup (landmark series)	Success
Crossing vessel present vs absent	**42% vs 86%**
High- vs low-grade hydronephrosis	60% vs 81%
Both excluded	95% (≈ pyeloplasty)

- Other series disagree → crossing vessel is **prognostic, not a contraindication**; but it pushes primary UPJO towards pyeloplasty.
- **Secondary UPJO — cross over**: failed endopyelotomy → pyeloplasty; failed pyeloplasty → endourology first (still succeeds often enough). Robotic redo is demanding; buccal mucosa graft and near-infrared fluorescence help.
- **Nephrectomy**: thin parenchymal shell + normal contralateral kidney. Adults **<15%** differential function = non-salvageable; children — no agreed cut-off. Uncertain → temporary stent/nephrostomy, re-scan. Also extensive stones with chronic infection, or multiple failed repairs — only if the other kidney is adequate.

4 · ENDOPYELOTOMY

- **Principle**: full-thickness **lateral** incision from lumen to peripelvic fat, healing over a stent. Lateral because that aspect is **free of crossing vessels**; direct vision lets any vessel be avoided.
- Pyeloplasty (≈**95%**) wins because it creates a wide, dependent funnel and reduces the pelvis; endourology wins only on recovery.
- **Contraindications**: segment >**2 cm**, active infection, uncorrected coagulopathy; marked vessel entanglement on three-dimensional (3D) CT. Such cases go robotic.
- Sterile urine before incision; if infection cannot clear because of obstruction → temporise with stent or nephrostomy.

Percutaneous antegrade

- Best for **UPJO + pelvicalyceal stones** (one sitting). Puncture **mid-posterior or superolateral calyx**, not lower pole (acute infundibular angle blocks passage across the UPJ); upper-pole access risks hydropneumothorax. No incision until a working + safety wire cross the UPJ.
- Tools: cold knife, electrocautery, holmium:yttrium-aluminium-garnet (Ho:YAG) or thulium fibre laser.
- High insertion: extend incision into the dependent pelvis; anterior insertion → incise truly posteriorly.
- Stent: 14/7-Fr endopyelotomy stent (wide end across UPJ), or 8–10 Fr if passage difficult; traditionally **6 weeks** (2 vs 4 weeks equal). No strenuous activity 8–10 days.
- **Clear stones before incising** — fragments migrate periureterally → fibrosis. A pelvic stone may cause oedematous pseudo-UPJO → clear, then reassess (Whitaker through the retained nephrostomy). Laparoscopic/robotic pyeloplasty + pyelolithotomy is equally effective.

Retrograde ureteroscopic (standard)

- Direct vision without a tract; most cost-effective. Contraindicated: long segment, coexisting stones. Narrow ureter → balloon dilate, or stent and return in 5–10 days (passive dilation).

ALGORITHM
General anaesthesia → wire + safety wire → flexible ureteroscope via sheath → 200-µm holmium fibre → incise **laterally** into fat, extending into normal ureter → stent + Foley catheter → renogram 4 weeks after stent removal

- Bleeding: haemostatic laser settings or balloon tamponade for 10 min. Excluding crossing vessels >4 mm raises success to ≈90%.
- Balloon dilation alone decays to ≈42%; blind cautery wire balloon → more haemorrhage and embolisation. Neither recommended.

- **Percutaneous endopyeloplasty**: endoscopic Heineke–Mikulicz repair; fails in secondary UPJO (scar).

Complications and follow-up

ALGORITHM
Percutaneous bleeding (thin atrophic parenchyma): bed rest, transfusion → **do not irrigate** nephrostomy (tamponade) → persistent → **selective embolisation** — low threshold, avoids exploration and nephrectomy

- Infection: first/second-generation cephalosporin prophylaxis (American Urological Association [AUA]). Clot in stent → a few more days of nephrostomy drainage. Retrograde route: mainly leak, stent migration, infection; ureteroarterial fistula reported.
- Follow-up: renogram 1 month after stent removal → 6 months → yearly for **2–3 years**. Durability gap widens: pyeloplasty 85/75% vs endopyelotomy 63/41% at 3/10 years.
- Failure → CT angiography; significant crossing vessel → **pyeloplasty, not repeat incision**.

5 · PYELOPLASTY

KEY POINT **Principles (open = minimally invasive)**: widely patent, watertight, tension-free anastomosis; funnel-shaped, **dependent** pelvis-to-ureter transition.

- MCQ Trendelenburg 1886 first attempt (patient died); Kuster 1891 first successful dismembered; **Anderson–Hynes 1949** = today's standard; Davis 1943 intubated ureterotomy; laparoscopic 1993; robotic 2002.

Dismembered (Anderson-Hynes)

- Almost universal: any insertion, pelvic reduction, **transposition over crossing vessels**. Only technique that excises the abnormal UPJ. Poorly suited to long proximal strictures or small intrarenal pelvis.
- Open approaches: extraperitoneal flank (default); anterior extraperitoneal (horseshoe/pelvic kidney); posterior lumbotomy (thin, no previous surgery).

ALGORITHM
Dissect ureter with periureteral tissue → lateral marking stitch → excise UPJ → spatulate **laterally** → apex to inferior pelvis → transpose anterior to vessel → watertight absorbable anastomosis (± reduction) → stent → drain

- **Stent** out at **4–6 weeks** — reduces extravasation and fibrosis. Added nephrostomy in uncomplicated adults → longer stay, more infection; reserve for secondary UPJO. Suction drain near, not on, the suture line.
- **Laparoscopic/robotic**: less morbidity, success >**90–95%** (= open). Robot's advantage is **ergonomic** (wristed suturing, short learning curve), not outcome. Transperitoneal most used (space, familiar anatomy); retroperitoneal smaller space; transmesenteric needs a visible dilated pelvis (pre-stenting hides it).
- **Absolute contraindications** to laparoscopy: uncorrected coagulopathy, untreated UTI, cardiopulmonary compromise. "Strict" success definitions (no pain + no obstruction on scan) are about one-third lower.
- Avoid extensive dissection and diathermy near the ureter. Antegrade stent: **clamp Foley** — urine from stent side-holes confirms distal coil in bladder.
- **Vascular hitch**: vessel fixed cranially without dismemberment (children); intrinsic UPJ disease left untreated.

ALGORITHM
Foley out 24–36 h → drain out if negligible; output rises after Foley removal → **replace Foley 7 days** (reflux along stent) → stent out 4–6 weeks → renography

- Complications: colonic injury, haemorrhage, ileus, urinoma, thromboembolism. Safe in infants and horseshoe/pelvic kidneys.
- Up to 30% of failures occur after 2 years → long follow-up. Failed pyeloplasty: endoscopy usually; redo pyeloplasty, ileal ureter, autotransplantation or nephrectomy.

6 · OTHER UPJ RECONSTRUCTIONS

> **KEY POINT** Every UPJ repair asks: **what tissue bridges the gap while keeping blood supply?** Dismemberment excises disease but devascularises; flaps keep blood supply but carry disease forward; the calyx is used when the pelvis is too small.

Procedure	Indication	Note
Foley Y-V-plasty	**High insertion**	Not with crossing vessels or need for reduction
Culp–DeWeerd spiral flap	Large pelvis, long proximal narrowing	Longest flap; length:width ≤**3:1**
Scardino–Prince vertical flap	Dependent UPJ, box-shaped pelvis	Shorter flap
Davis intubated ureterotomy	Long strictures, with spiral flap	Ureterotomy heals over stent (secondary intention)
Ureterocalicostomy	Small intrarenal pelvis, horseshoe, failed pyeloplasty	Ureter to lower-pole calyx

- **Y-V**: V-flap base on dependent medial pelvis, apex at UPJ; stem of Y down lateral ureter into normal calibre. Interrupted sutures (less pursing and ischaemia).
- Flaps: outline longer than needed (it shrinks once the pelvis opens); trim apex, never base (blood supply).
- **Intubated ureterotomy**: distal ureterotomy left open over a soft stent + nephrostomy; at 6 weeks nephrostogram → stent out → clamp → remove nephrostomy.
- **Ureterocalicostomy**: **resect lower-pole parenchyma** (simple nephrotomy → restricture) → mucosa-to-mucosa anastomosis → cover with fat/omentum. Cortex <5 mm predicts failure.

7 · RETROCAVAL URETER

- Persistent **posterior cardinal vein**; **always right**. **S-shaped deformity** (†"fish-hook") on pyelography; 3D CT definitive. Treat functionally significant obstruction only.
- **The anomaly is venous, so move the ureter**: divide, excise redundant segment, transpose anterior to the inferior vena cava (IVC), spatulated anastomosis over stent.

8 · URETERAL STRICTURE — EVALUATION AND ENDOUROLOGY

- **Causes**: malignancy, stone, radiation, surgical ischaemia, periureteral fibrosis (aneurysm, endometriosis), endoscopy, renal ablation, infection (tuberculosis, schistosomiasis), idiopathic.
- **Exclude malignancy first**: urothelial carcinoma (goblet sign) and metastases mimic benign stricture → CT ± ureteroscopy with biopsy.
- Stricture after ureteroscopy (URS) ≈**3%** vs 1.5% after shock-wave lithotripsy (SWL); impacted stone >2 months → ≈1 in 4. Thulium laser linked with more strictures than holmium (higher energy).
- Indications: exclude malignancy, ongoing obstruction, recurrent pyelonephritis, pain.
- Endourology needs ipsilateral function ≥**25%** (algorithm 20%).
- **Stents**: good for intrinsic strictures; extrinsic compression defeats a single stent → tandem stents or full-length metallic stent (changed yearly; encrustation, ingrowth). Chronic stenting suits patients unfit for repair or on cancer therapy.

- **Balloon dilation**: retrograde pyelogram → wire → 4-cm, 5–8-mm high-pressure balloon inflated until the **waist disappears**, 10 min → stent 2–4 weeks; antegrade if retrograde impossible. Works by longitudinal splits, like an incision. Contraindicated: infection, stricture **>2 cm**. Favourable: short, recent (≤3 months), non-anastomotic, well vascularised (≈85%); anastomotic ≈50%; transplant strictures respond well.
- **Endoureterotomy** (retrograde holmium, first line <2 cm): full thickness, including **2–3 mm normal ureter** each side; safety wire always; larger stents (8–12 Fr) better; triamcinolone injection may help.

Level	Incision	Avoids
Distal/mid ureter	**Anteromedial**	Iliac vessels
Proximal ureter	**Lateral/posterolateral**	Great vessels

- **Complete obliteration** (high-risk only): antegrade + retrograde scopes, one light off, **cut toward the light** → stent 8–10 weeks.

ALGORITHM
<2 cm + function >20% → endoscopic incision | >2 cm + >20% → reconstruction | <20% → nephrectomy (children may still merit repair). Transplant → consider balloon. Renography for 2 years

KEY POINT Each failure predictor (long, old, anastomotic, poorly vascularised) is a proxy for **ischaemic scar** that an incision cannot heal across.

9 · URETERAL STRICTURE — RECONSTRUCTION LADDER

Define nature, site, length; exclude carcinoma. Tension-free length is often judged only at surgery → be ready for the next rung. Iatrogenic injury → *Ch 94*.

Technique	Defect bridged	Level
Ureteroureterostomy	**2–3 cm**	Upper/mid
Ureteroneocystostomy	4–5 cm	Distal
Psoas hitch	**6–10 cm**	Below pelvic brim
Boari flap	**10–15 cm**	Lower/mid; spiral may reach pelvis
Renal descensus	5–8 cm	Upper; adds to any repair

- **Ureteroureterostomy**: short defects only (tension → stricture). Minimal handling, preserve adventitia; gunshot → excise devitalised ureter plus margin (blast effect); spatulate 5–6 mm 180° apart; double-J (methylene blue reflux from bladder confirms position); cover with fat/omentum. Drain out after 24–48 h minimal output (check drain creatinine). Success >90%.

ALGORITHM
Persistent leak: X-ray stent position → drain off anastomosis → stop suction → Foley + anticholinergic → nephrostomy

- **Psoas hitch/Boari**: contraindicated by small contracted bladder → urodynamics; treat outlet obstruction first. Boari's commonest complication: **recurrent stricture** (ischaemia/tension). Refluxing = non-refluxing reimplant in adults. Robotic reimplant = open success, shorter stay. eGFR (estimated glomerular filtration rate) fell in 44% within 2 years despite resolved hydronephrosis — counsel.
- **Renal descensus**: mobilise kidney in Gerota fascia, rotate inferomedially, fix lower pole; limited by the renal vein.
- **Transureteroureterostomy (TUU)**: contraindicated by short donor ureter, diseased recipient, or disease affecting both ureters (stones, RPF, urothelial cancer, radiation). Donor tunnelled **above**

the inferior mesenteric artery (IMA) (prevents kinking); correct recipient reflux (voiding cystogram); minimal recipient mobilisation, end-to-side.

- **Ileal ureter**: isoperistaltic; contraindicated by creatinine >**2 mg/dL** (acidosis in ≈half), bladder dysfunction, inflammatory bowel disease, radiation enteritis. Distal ileum ≥15 cm from ileocaecal valve; segments >15 cm prevent reflux to the pelvis. No proven benefit of tapering. Complications: leak, mucus obstruction, ischaemic necrosis. Progressive dilation + worsening acidosis → seek outlet obstruction; endoscopic surveillance from year 3.
- **Buccal mucosa onlay** (robotic, omental wrap): ureter encased in dense scar. **Autotransplantation**: contralateral kidney absent/poor or other repairs impossible.

> **KEY POINT** Ladder: bring the bladder up, the kidney down, borrow the other ureter, or substitute (ileum, buccal). Tension and ischaemia cause failure at every rung.

10 · URETEROENTERIC ANASTOMOTIC STRICTURE

- Conduits 4–9%; ≈5% after 10 years → **lifelong upper-tract surveillance**.
- **Cause = ischaemia**: blood supply runs in the adventitia. **Left commoner**: extra length, passage under sigmoid mesentery, angulation at IMA.
- Risk: **non-refluxing anastomosis** (≈13% vs 2–3%; reflux into a low-pressure reservoir is not clearly harmful, obstruction is), carcinoma in situ at margin, obesity, American Society of Anesthesiologists (ASA) score >2, Clavien–Dindo 3–4 complications. Wallace and Bricker both acceptable.
- Conduit dilation is often non-obstructive → confirm with renography. Irregular mass + rapid progression → recurrent cancer.
- **Open revision = gold standard** (≈91% vs 26% endoscopic); robotic equal. Endourology is **antegrade** (nephrostomy usually present) → balloon or holmium incision → internal–external stent (flushable — mucus plugs). Success <1 cm ≈50% vs >1 cm 6%; right ≈41% vs left 19%.

ALGORITHM
<2 cm + >20% → right: endoscopic incision; **left: formal repair** (sigmoid mesentery bleeding) | >2 cm → formal repair | <20% → nephrectomy or observation

11 · RETROPERITONEAL FIBROSIS

- Age 40–60, male:female 2–3:1. Mass centres on distal aorta at **L4–L5**, encasing ureters; usually bilateral. Back/flank pain commonest; weight loss, fever, hypertension; half present uraemic. IVC involvement → leg oedema, deep vein thrombosis.

Idiopathic (≈70%)	Secondary (≈30%)
Chronic periaortitis (large-vessel vasculitis); ceroid antigen from atherosclerotic plaque	**Drugs (commonest identifiable)**: methysergide/ergots, β-blockers, phenacetin
Multifocal fibrosclerosis: sclerosing mediastinitis/cholangitis, orbital pseudotumour, Riedel thyroiditis	Malignancy: **lymphoma** (commonest), myeloma, carcinoid, sarcoma
Autoimmune; antinuclear antibody predicts relapse. †Immunoglobulin G4 (IgG4)-related disease is a recognised subset — check serum IgG4	Radiotherapy, asbestos, infection (tuberculosis)

- Malignant RPF shows only small tumour islands → **deep, multiple biopsies**. Raised erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP).
- **Imaging**: **medial deviation** of mid-ureters; magnetic resonance imaging (MRI) high T2 = active; positron emission tomography (PET) most sensitive for activity, may obviate biopsy, used for follow-up. Biopsy can be omitted if no malignancy history, atypical imaging or lymphadenopathy.

ALGORITHM
Uraemia → **nephrostomy** or stents → manage **post-obstructive diuresis** → stop culprit drug, biopsy → steroids → ureterolysis

- **Steroids first line** (prednisone 60 mg tapering, up to 2 years; ≈80% respond); second line immunosuppressants, **tamoxifen**. Steroids act while the lesion is cellular: pain and ESR fall within days. Relapse on taper 25–50%. Only ≈15% need ureterolysis.
- **Ureterolysis — bilateral**; start at distal non-dilated ureter; "split and roll". Then **isolate ureter**: intraperitonealisation or **omental wrap** on gastroepiploic pedicles (most definitive — barrier plus blood supply). Post-op steroids may prevent recurrence. Ureterolysis impossible → autotransplantation; dead kidney + good contralateral → nephrectomy.

12 · MCQ ONE-LINERS

1. Obstructed renogram: T½ >40 minutes despite furosemide.
2. Whitaker: 10 mL/min; obstructed >15–20 cm H_2O.
3. Endopyelotomy with crossing vessel: 42% vs 86% without.
4. Endopyelotomy incision: lateral.
5. Avoid lower-pole access for antegrade endopyelotomy.
6. Adult kidney <15% function: non-salvageable.
7. Standard pyeloplasty: Anderson–Hynes (1949).
8. Foley Y-V-plasty: high insertion only.
9. Spiral flap length:width ≤3:1.
0. Retrocaval ureter: always right-sided.
1. Endoureterotomy: distal anteromedial, proximal lateral.
2. Commonest Boari complication: recurrent stricture.
3. TUU: donor ureter passes above the IMA.
4. Ileal ureter contraindicated with creatinine >2 mg/dL.
5. Ureteroenteric stricture commoner on left.
6. RPF: aorta at L4–L5; medial ureteric deviation.
7. RPF first line: prednisone.
18. Ureterolysis: always bilateral.
19. PET: most sensitive test of RPF activity.
20. Commonest identifiable RPF cause: drugs (methysergide); commonest malignant: lymphoma.
21. Furosemide in renography: 0.3–0.5 mg/kg at 20 minutes.
22. Preferred renography isotope: MAG3.
23. Crossing vessel: prognostic factor, not contraindication, for endopyelotomy.
24. Endourological contraindication: obstructed segment >2 cm.
25. Only dismembered pyeloplasty excises the abnormal UPJ.
26. Ureterocalicostomy: resect lower-pole parenchyma, not simple nephrotomy.
27. Stricture after URS ≈3% vs SWL 1.5%.
28. Psoas hitch bridges 6–10 cm; Boari flap 10–15 cm.
29. Left ureteroenteric stricture: formal repair (sigmoid mesentery bleeding).
30. Non-refluxing ureteroenteric anastomosis strictures more often.
31. Ureterolysis: start distally; omental wrap most definitive.

94 Upper Urinary Tract Trauma

Reading time ≈ 35 min (slow pace)

1 · RENAL INJURY — MECHANISM AND PRESENTATION

- Kidney is the **commonest urological organ injured** by external trauma. Blunt 80–90% (motor vehicle collision commonest); firearms cause ~⅔ of penetrating injuries. Most are grade I–III and heal conservatively.
- **Deceleration** injures the kidney at its fixation points — hilum and ureteropelvic junction (UPJ): renal artery thrombosis, vein disruption, pedicle avulsion, UPJ disruption. Ask vehicle speed and fall height.
- **Penetrating**: more bleeding, nephrectomy and complications than blunt. High-velocity missiles → blast and cavitation → delayed necrosis. Stab at anterior axillary line → hilum/pedicle; posterior axillary line → parenchyma.
- Signs: flank ecchymosis or tenderness, **ipsilateral rib 9–12 fractures** (strongest association), low thoracic/flank wound. Mark gunshot entry/exit with metallic markers on plain film to define trajectory.

2 · HAEMATURIA

- Significant injury suggested by **gross haematuria**, or **microhaematuria (>5 red blood cells per high-power field [RBC/hpf]) with shock** (systolic blood pressure [SBP] **<90 mm Hg**).
- Degree does not correlate with severity; absent in some grade IV injuries, ~⅓ of blunt renovascular and **~½ of UPJ injuries**.
- Test the **first urine aliquot** (later samples are diluted by resuscitation). Microscopy is gold standard; dipstick false positives: myoglobinuria, menses, povidone-iodine.

> **KEY POINT** Haematuria triggers imaging; it does not measure injury. The worst injuries (pedicle, UPJ) may not bleed into the urine, so mechanism carries equal weight.

3 · CLASSIFICATION — AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA (AAST) KIDNEY INJURY SCALE (2018)

The American Association for the Surgery of Trauma (AAST) scale is graded on contrast-enhanced computed tomography (CT) and predicts intervention and nephrectomy. †The 2018 revision is the current version.

Grade	AAST 2018
I	Subcapsular haematoma and/or parenchymal contusion, no laceration
II	Perirenal haematoma confined to Gerota's fascia; laceration **≤1 cm**, no extravasation
III	Laceration >1 cm, no collecting-system rupture; **any vascular injury or active bleeding contained within Gerota's fascia**
IV	Laceration into collecting system with extravasation; renal pelvis laceration or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's fascia; segmental or complete infarction from thrombosis without active bleeding
V	**Main renal artery or vein laceration** or hilar avulsion; devascularised kidney with active bleeding; shattered kidney with loss of identifiable anatomy

- Advance one grade for bilateral or multiple injuries up to grade III; grade by the highest finding on imaging, operation or pathology.
- Grades I–III defined by **depth (1 cm)**; IV–V by collecting system and vessels. MCQ Trap: main-vessel injury with contained bleeding was grade IV in 1989, now V; segmental vascular injury (pseudoaneurysm, arteriovenous fistula) now IV.

4 · INDICATIONS FOR IMAGING

American Urological Association (AUA) and European Association of Urology (EAU) guidelines agree:

- Penetrating trauma — **any degree of haematuria** (pedicle injury occurs with minimal microhaematuria), if stable enough for CT.
- Blunt trauma with rapid deceleration, gross haematuria, or microhaematuria with SBP <90 mm Hg at any time.
- Blunt microhaematuria without shock or deceleration → **observe without imaging** unless examination suggests injury (San Francisco selective-imaging series).
- **Children**: higher risk (relatively larger kidney, less fat, pliable ribs); catecholamines maintain blood pressure (BP) until ~50% blood volume lost — hypotension is late, so image liberally. Congenital anomalies (UPJ obstruction) let minor trauma cause major injury.

5 · IMAGING STUDIES

- **Contrast CT with arterial/nephrographic and delayed excretory phases** is gold standard: arterial phase shows bleeding; delayed phase (**~10 min**) shows urinary extravasation, which is missed without it. CT cannot reliably show **renal vein injury** — perfusion looks normal; medial haematoma with normal arterial findings suggests it.
- Delayed films may be omitted if kidneys are normal with no perinephric or pelvic fluid; if collecting-system injury is suspected and no delay was obtained, repeat a urographic phase.

CT finding	Suggests
Medial haematoma	Vascular pedicle injury
Medial extravasation; no contrast in distal ureter	Renal pelvis/UPJ avulsion
Absent global enhancement (cortical rim sign)	Main renal artery occlusion
Intravascular contrast extravasation (blush)	Brisk bleeding → strongly predicts angioembolisation (AE)
≥2 of: haematoma rim distance >4 cm, medial laceration, contrast blush	High-risk grade IV

- **Multi-Institutional Genito-Urinary Trauma Study (MiGUTS) nomogram** predicts bleeding intervention: penetrating mechanism, contrast blush, pararenal haematoma, associated injuries, shock and — largest contributor — **haematoma rim distance**.
- **Intraoperative one-shot intravenous pyelogram (IVP)**: unstable patient at laparotomy without CT, mainly to confirm a **functioning contralateral kidney**: **2 mL/kg contrast → single film at 10 min**. Alternatives: palpate the other kidney, or loop the injured ureter and give intravenous (IV) methylene blue.
- Focused assessment with sonography for trauma (FAST): poor specificity; cannot separate haematoma from urine.

Perinephric collection	Hounsfield units (HU)	Feature
Urinoma	0–20	Contrast pooling on delayed phase
Haematoma	>30	—
Abscess	>20	Rim enhancement

6 · NONOPERATIVE MANAGEMENT

- Stable, accurately staged **grade I–IV and selected grade V** → expectant management regardless of mechanism; >90% success in blunt injury even with extravasation.
- Protocol: vital signs, haemoglobin every 4–6 h, serial examination. Bed rest is empirical; **early mobilisation is safe**.

- **Failure triggers**: instability; transfusion of **2–3 units**; new flank pain or guarding; enlarging urinoma with fever or rising white count. Most failures need only stent, nephrostomy or AE.
- **Penetrating** injury breaches Gerota's fascia, losing the tamponade blunt injury preserves, so more fail and associated intra-abdominal injury more often forces laparotomy. Stab wounds do better than gunshot; isolated stable stab posterior to the posterior axillary line → serial examination; new tenderness or guarding → explore.
- Failure within 24 h is rare (~3%); risk factors: higher grade, penetrating mechanism, nonrenal abdominal injury.
- **Reimaging**: AUA — consider CT at ~48 h for **grade IV–V with deep lacerations** or clinical deterioration; not for uncomplicated I–III. Delayed bleeding after discharge is rare, usually within 2–3 weeks, and embolised.

7 · URINARY EXTRAVASATION

- Parenchymal leaks: **76–90% heal spontaneously**.
- **Drain early** (stent ± nephrostomy) when urine cannot pass antegrade: pelvis/UPJ avulsion, ureteric clot, UPJ obstruction, parapelvic cyst.
- Symptomatic leak or enlarging urinoma → retrograde stent + urethral catheter → add percutaneous nephrostomy or urinoma drain if persistent.
- Retrograde stent impossible (pelvic fracture urethral injury, ureteral transection, no lithotomy) → percutaneous nephrostomy ± antegrade stent.
- >25% nonviable parenchyma with extravasation → more abscess; lower threshold to operate.

> **KEY POINT** Parenchymal leaks heal because antegrade flow is preserved; anything stopping urine reaching the bladder converts an observable leak into one that must be drained.

8 · INTERVENTION

Absolute indications	Relative (increasingly observed)
Instability with no or transient response to resuscitation	Extravasation with significant devascularisation
Expanding or pulsatile haematoma (renal artery laceration)	Associated colon or pancreatic injury
Suspected pedicle avulsion; UPJ avulsion	Arterial thrombosis; parenchymal extravasation

Angioembolisation

- Superselective AE stops bleeding without laparotomy and raises salvage; repeat AE is highly successful. Needs an experienced team, no delay and resuscitation in the suite.
- **Failure predictors**: instability, grade V, urinary extravasation, gross haematuria. Late pseudoaneurysm or arteriovenous fistula (AVF) → AE.

Exploration and damage control

- At laparotomy for other injuries, exploration is **not mandatory** for staged grade I–III; only absolute indication is **pulsatile or expanding haematoma**. Opening Gerota's fascia otherwise releases tamponade → avoidable nephrectomy.
- Transabdominal approach; principle — **isolate renal vessels before opening Gerota's fascia**.

ALGORITHM
Eviscerate small bowel → incise peritoneum over aorta above inferior mesenteric artery → extend to ligament of Treitz → follow aorta to left renal vein → loop vein, retract cephalad → loop both renal arteries → reflect colon → open Gerota's fascia → occlude vessels if bleeding

- **Damage control** (hypothermia, acidosis, coagulopathy): control haemorrhage → pack → temporary closure → intensive care resuscitation → return at 24–48 h; may permit AE instead of nephrectomy.
- **Nephrectomy**: decided by stability, reconstructability, associated injuries and a **functioning contralateral kidney**.

Renal reconstruction

ALGORITHM
Exposure → vascular control → limited debridement → suture-ligate vessels → **watertight collecting-system closure** → approximate parenchyma over bolster → cover with Gerota's fascia or **pedicled omentum** → drain

- Polar injury → partial nephrectomy with omental cover. Colon or pancreatic injury no longer mandates nephrectomy.

Renovascular injury

- Blunt artery thrombosis: deceleration stretches the artery → inelastic intima tears → thrombus.
- **Absolute indication for revascularisation**: solitary kidney or bilateral injury, warm ischaemia ideally **<120 min**. Unilateral with normal other kidney → observe (thrombosis) or nephrectomy; revascularisation salvages only ~⅓.
- **Left renal vein may be ligated** proximal to gonadal and adrenal collaterals; **right cannot**. Hypertension follows thrombosis repair in ≥43% → monitor BP for a year.
- Endovascular stenting of intimal-flap thrombosis is promising; drawback is anticoagulation in polytrauma.

9 · COMPLICATIONS AND FOLLOW-UP

- Mortality reflects associated injuries. Acute kidney injury relates to age, transfusion and polytrauma rather than nephrectomy itself.
- **Persistent leak/abscess**: stent → nephrostomy or percutaneous drain; open surgery rarely.
- **Delayed bleeding**: usually within 21 days → AE.
- **Hypertension**: uncommon; commoner after artery injury; partial ischaemia → renin–angiotensin activation.

Entity	Mechanism	Management
Goldblatt kidney	Artery stenosis or occlusion	—
Page kidney	Compression by subcapsular blood/urine or scar	Angiotensin-converting enzyme inhibitor or receptor blocker; refractory → drainage or decortication
Post-traumatic AVF	Shunt causing distal ischaemia	AE

- Follow-up: EAU — no imaging for grade I–II; salvaged III–V → dimercaptosuccinic acid (DMSA) scan. Minimum: blood pressure, urinalysis, renal function.
- Return to sport 2 weeks–3 months; secondary bleeding risk peaks **2–6 weeks**. A solitary kidney does not bar contact sport.

10 · URETERAL INJURY — AETIOLOGY AND GRADING

- **Iatrogenic is commoner overall** (gynaecological, colorectal, ureteroscopy). External: mostly gunshot; >90% have associated injuries; proximal ureter commonest external site. Blunt injury needs extreme deceleration (lumbar transverse process fractures).
- **Haematuria absent in 25–45%** — keep a high index of suspicion.

AAST ureter grade	Description
I	Contusion or haematoma without devascularisation
II	Laceration **<50%** transection
III	Laceration ≥50% transection
IV	Complete transection with **<2 cm** devascularisation

AAST ureter grade	Description
V	Avulsion with >2 cm devascularisation

Advance one grade for bilateral injury up to III. Trap: IV vs V is length of devascularisation, not completeness.

Iatrogenic injury

- **Distal ureter** commonest site; minimally invasive hysterectomy riskier than open; prolapse surgery highest gynaecological risk; endometriosis raises risk.
- In ~10% of women distal supply comes mainly from uterine artery branches → **devascularisation** presenting at 1 week–2 months as stenosis, urinoma or ureterovaginal fistula.
- **Ureteroscopy**: stricture ~3%, avulsion <1%. Stop basketing when a tear is seen and stent; holmium laser safest lithotripter; use ≤14-Fr sheaths in unstented ureters; work over a safety wire. Risk factors: long operating time, impacted or proximal stone, inexperience, prior radiation.
- Partial nephrectomy or ablation of lower-pole/hilar tumours → proximal stricture months later; pyeloperfusion protects during ablation.
- **Ureteroarterial fistula**: haematuria with chronic stent plus pelvic surgery or radiation → endovascular covered stent.

Prevention and detection

- Preoperative stents aid **recognition**, not proven prevention; risk anuria (bilateral), infection. Intraureteral indocyanine green fluoresces for hours.
- Cystoscopy with efflux check after minimally invasive hysterectomy and prolapse surgery. Efflux aids: oral phenazopyridine (most cost-effective), IV indigo carmine (safe), fluorescein, dextrose irrigation.
- **Methylene blue** is a monoamine oxidase inhibitor: serotonin toxicity with antidepressants; avoid in pregnancy and glucose-6-phosphate dehydrogenase deficiency; falsely lowers oximetry.

11 · URETERAL INJURY — DIAGNOSIS

Modality	Role
CT urography with **10-min delays**	Best study in stable patient; trace both ureters
Retrograde ureterography	**Most sensitive**; mandatory if suspicion persists; allows stenting
Antegrade study	When retrograde fails
One-shot IVP	Often non-diagnostic — never rely on it

- CT signs: **no contrast distal to injury** (transection), medial extravasation (UPJ avulsion), delayed nephrogram, periureteral urinoma.
- ~11% missed at laparotomy; missed injury presents with fever, leucocytosis, peritonism.

> **KEY POINT** Ureteral injury is diagnosed by absent contrast beyond it — so time the study for excretion, follow both ureters, and never treat a non-diagnostic study as negative.

12 · URETERAL INJURY — MANAGEMENT

ALGORITHM
Stable at laparotomy → repair now · unstable → damage control, delayed repair · contusion → stent or excise and repair · late or partial → stent → fails → nephrostomy, delayed repair · ureterovaginal fistula → stent first

- Principles: spare adventitia; debride to bleeding edges; **spatulated, tension-free, stented, watertight** anastomosis; drain; retroperitonealise. **No bowel or autotransplantation acutely.**
- Close a longitudinal laceration transversely (Heineke–Mikulicz) to avoid narrowing; omental wrap for blast or concomitant vascular repair.

- **Damage control** (shock, uncontrolled bleeding, colectomy): ligate proximal stump with long silk (aids later identification) → percutaneous nephrostomy once stable; or exteriorised single-J stent/5-Fr feeding tube. Open nephrostomy is too slow.

Level	Acute repair	Delayed / long loss
Upper	Ureteropyelostomy; ureteroureterostomy; ureterocalicostomy	Buccal graft; appendix (right); ileal
Mid	Ureteroureterostomy ± downward nephropexy	Transureteroureterostomy (TUU); ileal ureter
Distal	Ureteroneocystostomy ± **psoas hitch**	Boari flap

- **Ureteroneocystostomy** in trauma: **non-tunnelled (refluxing)** — tunnelled ones stricture more; reflux is harmless.
- **Psoas hitch**: mainstay for lower third (95–100%); preferred over ureteroureterostomy distally because the distal stump's blood supply is tenuous. Suture bladder to psoas minor tendon along the muscle length; avoid genitofemoral nerve entrapment and deep bites (femoral nerve); may divide the contralateral superior vesical pedicle.
- **Boari flap**: gaps up to **12 cm**; needs normal bladder; length:width ≤3:1; rarely acute.
- **Ureteroureterostomy**: commonest acute repair (upper two-thirds, above iliac vessels), up to 90% success; complications — leak 10–24%, urinoma, stricture up to ~12%.
- **Ureterocalicostomy**: ureter to exposed lower calyx after lower-pole partial nephrectomy, for profound pelvis/UPJ damage. UPJ avulsion → spatulated reimplant into the renal pelvis.
- **TUU**: injured ureter crosses midline, end-to-side into contralateral ureter; 90–97% success; delayed use only. Avoid with stones, urothelial cancer, tuberculosis, radiation or retroperitoneal fibrosis — risks converting a unilateral into a bilateral injury.
- **Ileal ureter** (delayed only): long-segment loss; metabolic acidosis, mucus; Yang–Monti reconfigured tube uses less bowel. **Autotransplantation** is the last resort before nephrectomy.
- Follow-up: renogram or ultrasound at 4–12 weeks and 1 year; late stenosis may be silent.

13 · MCQ ONE-LINERS

1. Ipsilateral rib 9–12 fractures: strongest association with renal injury.
2. Blunt microhaematuria without shock or deceleration: no imaging.
3. About half of UPJ injuries have no haematuria.
4. Children: hypotension only after ~50% blood loss.
5. AAST 2018: segmental vascular injury is grade IV.
6. AAST 2018: main renal vessel laceration is grade V.
7. Cortical rim sign: main renal artery occlusion.
8. Largest MiGUTS contributor: haematoma rim distance.
9. One-shot IVP: 2 mL/kg, film at 10 minutes.
0. Isolate renal vessels before opening Gerota's fascia.
1. Left renal vein can be ligated; right cannot.
2. Urinoma: 0–20 HU.
3. Page kidney: compression → renin-mediated hypertension.
4. Most sensitive test for ureteral injury: retrograde ureterography.
5. AAST ureter IV vs V: devascularisation <2 vs >2 cm.
16. Trauma ureteroneocystostomy: non-tunnelled.
17. Boari flap bridges up to 12 cm.
18. Methylene blue: serotonin syndrome risk.
19. Ureteroarterial fistula: endovascular covered stent.
20. Commonest site of iatrogenic ureteral injury: distal ureter.
21. Commonest site of external ureteral injury: proximal ureter.
22. Revascularisation absolute indication: solitary kidney or bilateral injury.
23. Only absolute intraoperative exploration indication: pulsatile or expanding haematoma.
24. AE failure predictors: instability, grade V, extravasation, gross haematuria.
25. Penetrating renal injury: image with any haematuria.
26. Delayed renal bleeding: usually within 21 days; treat by AE.
27. No bowel interposition or autotransplantation in acute ureteral repair.
28. Psoas hitch: lower-third mainstay, 95–100% success.

95 Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

Reading time ≈ 57 min (slow pace)

1 · OVERVIEW AND EPIDEMIOLOGY

- Stone treatment is two problems: **removal is surgical, recurrence is metabolic**. Prophylaxis → *Ch 99*.
- Lifetime prevalence **1–15%**; United States (US) prevalence ≈ **10%** and rising — partly a detection artefact of more computed tomography (CT).
- **Sex**: historically male 2–3 : 1, but the gap is narrowing (female incidence rising, steepest in adolescents). **Oestrogen is protective** — it enhances renal calcium reabsorption and reduces bone resorption; hence a second female peak at menopause.
- **Race**: highest in White, lowest in Asian individuals.
- **Age**: adult peak 4th–6th decades; paediatric stones rising 5–10%/year, peak in adolescent females.
- **Climate**: hot, arid regions; US "stone belt" in the Southeast; summer peak (July–September). Heat acts via **low urine volume**, sunlight via **vitamin D-mediated hypercalciuria**
- **Occupation**: heat (steel, glass workers) → low volume and pH → uric acid stones; sedentary work and weightlessness → bone turnover → hypercalciuria; restricted bathroom access → low volume.

Systemic associations

- **Obesity/body mass index (BMI)**: independent risk factor, stronger in females.
- **Metabolic syndrome**: risk rises with the number of traits. **Diabetes**: bidirectional — diabetes predicts stones in females; a stone history predicts diabetes in both sexes.
- **Unifying mechanism — insulin resistance**: low urine pH → uric acid stones; hyperinsulinaemia → hypercalciuria → calcium stones.
- **Hypertension, myocardial infarction (MI), stroke, chronic kidney disease (CKD) and end-stage renal disease (ESRD)**: associated with stones, mostly stronger in females, causation unproven — probably a shared metabolic soil.
- **Water**: fluid intake is inversely related to stones — proven by randomised trial (see trials). Water hardness: weak, conflicting effect.

2 · PHYSICOCHEMISTRY — STATE OF SATURATION

- Cascade: **supersaturation → nucleation → crystal growth/aggregation → retention at an anchoring site → stone**.
- **Concentration product (CP)**: product of the concentrations of a salt's components. **Solubility product (Ksp)**: CP at saturation (equilibrium). **Formation product (Kf)**: CP above which crystals form spontaneously.

State	CP	What happens
Undersaturated	< Ksp	No crystals; existing stones may dissolve
Metastable	Ksp–Kf (most urinary salts sit here)	No spontaneous (homogeneous) nucleation; growth on existing crystals and heterogeneous nucleation possible; **inhibitors effective — the zone of therapeutic intervention**
Unstable	> Kf	Spontaneous nucleation, growth, aggregation; inhibitors ineffective

- Crystals still form in metastable urine because (1) local CP exceeds Kf in parts of the nephron, (2) obstruction/stasis prolongs transit, (3) **heterogeneous nucleation** on impurities needs far less energy than homogeneous nucleation.
- **Relative saturation ratio (RSR)** = urine CP ÷ Ksp (computer program EQUIL 2): < 1 dissolve, > 1 form and grow. EQUIL 2 overestimates; the Joint Expert Speciation System (JESS) agrees with

direct measurement.

- **Lowering saturation**: dilute; lower calcium or oxalate; complex calcium with citrate; change pH — works for **phosphate, uric acid and cystine but not oxalate** (oxalic acid is a strong acid, pK ≈ 4, so physiological pH barely changes it).
- **Calcium and oxalate contribute equally** to calcium oxalate saturation → lowering either helps.

3 · NUCLEATION, GROWTH, AGGREGATION AND RETENTION

- Normal urine holds calcium oxalate at ≈ 4× its aqueous solubility; with inhibitors it precipitates only at **7–11×**.
- In vivo nucleation is usually **heterogeneous** — on epithelial cells, debris or other crystals.
- Nephron transit ≈ **5–7 min** — too short for a single crystal to occlude a tubule, but **aggregation** of many nuclei can do so within minutes.
- **Free-particle theory**: crystals grow free in tubular fluid and are trapped (ductal plugs) — recalculation shows it is possible. **Fixed-particle theory**: crystals bind an anchoring site (Randall plaque), prolonging exposure to supersaturated urine.

Randall plaque

- Subepithelial papillary plaque (Randall, 1937 [MCQ]) — present in most stone formers on endoscopy.
- **Origin: basement membrane of the thin limbs of the loop of Henle** → medullary interstitium → subepithelial papilla → erodes through urothelium → exposed surface on which **calcium oxalate grows attached**.
- Plaque mineral is **calcium apatite** (carbapatite), not calcium oxalate; laminated mineral–organic layers with osteopontin at the interface.
- Vascular hypothesis: vasa recta injury → atherosclerosis-like calcification.

Stone former	Randall plaque	Ductal plugs
Idiopathic calcium oxalate	Present — anchoring site	Uncommon (up to a third, mostly mild)
Intestinal bypass (enteric hyperoxaluria)	Absent	Apatite plugs in inner medullary collecting ducts; epithelial injury, fibrosis
Brushite	Present	Apatite plugs; duct injury, interstitial fibrosis
Distal renal tubular acidosis (RTA), hyperparathyroidism, cystinuria	—	Present

> KEY POINT **Plaque vs plug**: plaque is the idiopathic route (an interstitial apatite anchor); plug is intratubular crystal obstruction with tubular injury — the route of secondary systemic disease. **Ductal plugs should prompt a search for a secondary cause.** Insults such as infection or shock-wave lithotripsy (SWL) may convert a calcium oxalate former into a brushite former.

4 · INHIBITORS, PROMOTERS AND MATRIX

- Urine is normally supersaturated for calcium oxalate and phosphate; stones do not form because inhibitors raise the supersaturation needed for nucleation and slow growth/aggregation. **No inhibitor of uric acid (or cystine) is known** — these stones depend on pH and concentration alone.
- Promoters of calcium oxalate supersaturation: low volume, low citrate; high calcium, oxalate, phosphate, uric acid.

Inhibitor	Action	Key feature
Citrate	Complexes calcium; inhibits calcium oxalate precipitation, agglomeration and urate-induced heterogeneous nucleation	The most important small-molecule inhibitor; potent against calcium phosphate growth
Magnesium	Complexes oxalate	Synergy with citrate and phytate
Glycosaminoglycans	Bind crystal surface calcium	Lower in (recurrent) stone formers
Nephrocalcin	Inhibits calcium oxalate monohydrate nucleation, growth, aggregation	Stone formers excrete non-inhibitory isoforms lacking γ-carboxyglutamic acid
Tamm-Horsfall protein (uromodulin)	Inhibits aggregation, not growth	Most abundant urinary protein; from thick ascending limb (TAL); inhibitor in alkaline urine but **promoter (polymerises) in acid urine**
Osteopontin (uropontin)	Inhibits nucleation, growth, aggregation and crystal binding to epithelium	Inducible inhibitor

- **Matrix** = non-crystalline component, ≈ **2.5%** of stone weight; contains **matrix substance A** (in urine of stone formers, not healthy people) and mostly inflammatory proteins. Role undetermined.

5 · MINERAL METABOLISM

Calcium

- 30–40% of dietary calcium absorbed, mainly small intestine (colon ≈ 10%). **Transcellular**: saturable, calcitriol-dependent, dominates at low intake. **Paracellular**: non-saturable, dominates at high intake.
- Only ionised calcium is absorbed; luminal phosphate, citrate, oxalate, sulphate and fatty acids bind it. So **oxalate-rich food lowers calcium absorption and luminal calcium lowers oxalate absorption**.
- **Vitamin D**: skin 7-dehydrocholesterol → (sunlight) previtamin D3 → liver 25-hydroxylation → proximal tubule **1α-hydroxylation** → 1,25-dihydroxyvitamin D3 (calcitriol), the most potent stimulator of gut calcium absorption. 1α-hydroxylase is stimulated by **parathyroid hormone (PTH) and hypophosphataemia**.
- **PTH** (secreted when the calcium-sensing receptor [CaSR] senses low serum calcium): bone resorption; kidney ↑ calcium reabsorption, ↓ phosphate reabsorption, ↑ calcitriol; **no direct gut effect**.

Segment	Filtered calcium reabsorbed	Route / molecules
Proximal tubule	**60–65%**	Paracellular; claudin-2
TAL	25–30%	Paracellular, lumen-positive voltage; claudin-16/19 channel; CaSR → claudin-14 blocks it
Distal convoluted tubule (DCT)	8–10% (fine control)	Transcellular: transient receptor potential vanilloid 5 (TRPV5) → calbindin-D28k → basolateral exit

- Only 1–3% of filtered calcium is excreted.

Oxalate

- Only 6–14% of dietary oxalate absorbed, roughly half in small bowel, half in colon; **not regulated** and almost entirely excreted in urine — urinary oxalate tracks what the gut lets through plus hepatic synthesis.
- Urinary oxalate ≈ half dietary, half endogenous (liver; glyoxylate is the direct precursor). **Ascorbic acid** is the largest endogenous source (up to 40%); hydroxyproline (collagen, gelatin) ≈ 15%.

- Modifiers: luminal calcium/magnesium bind oxalate; *Oxalobacter formigenes* degrades oxalate (energy source) and stimulates colonic oxalate secretion. Small-bowel disease with an intact colon → marked hyperabsorption.

6 · CLASSIFICATION OF NEPHROLITHIASIS

Stone	Frequency
Calcium oxalate (pure or mixed with apatite)	**60%**
Hydroxyapatite (mostly mixed with calcium oxalate)	20%
Brushite	2%
Uric acid	≈ 7% (8–10% US, rising)
Struvite (infection)	≈ 7% (falling)
Cystine	1–3% (up to 10% in children)
Triamterene, silica, 2,8-dihydroxyadenine	< 1% each

Condition	Defect	% of formers
Absorptive hypercalciuria (incl. renal phosphate leak)	↑ Gut calcium absorption	20–40
Renal hypercalciuria	Renal calcium leak	5–8
Resorptive hypercalciuria	Primary hyperparathyroidism	3–5
Hyperuricosuric calcium	Purine excess, urate overproduction	10–40
Hypocitraturic calcium	Idiopathic, chronic diarrhoea, distal RTA, thiazide (hypokalaemia)	10–50
Hyperoxaluric calcium	Primary, enteric, dietary	2–15
Hypomagnesiuric calcium	↓ Gut magnesium absorption	5–10
Idiopathic low urine pH	Low urinary pH	15–30
Low urine volume	Inadequate intake	10–50
Infection; cystinuria	Urease bacteria; cystine reabsorption defect	1–5; < 1
No abnormality	—	< 3

KEY POINT Classification is by the **urinary abnormality, not the stone**: calcium stones are a final common product of many derangements, so composition starts the work-up rather than ending it. Fewer than 3% of formers have no identifiable abnormality — evaluation almost always finds a target. Work-up and treatment → *Ch 99*.

7 · HYPERCALCIURIA

- Commonest abnormality in calcium stone formers (35–65%).
- Mechanism: ↑ saturation of calcium salts and ↓ inhibitor activity (calcium complexes citrate and chondroitin sulphate).
- **Definition**: > **200 mg/day** after 1 week of 400 mg calcium, 100 mEq sodium diet; or > 4 mg/kg/day; or > 7 mmol/day (men), > 6 mmol/day (women). Any cut-off is artificial — risk is graded.
- **Pak classification** (by organ: gut, kidney, bone) — the standard exam framework, though not routinely used to direct therapy.

Feature	Absorptive	Renal	Resorptive
Defect	↑ Gut calcium absorption	Renal calcium leak	Primary hyperparathyroidism (PHPT)

Feature	Absorptive	Renal	Resorptive
Serum calcium	Normal	Normal	**High**
PTH	Normal / **suppressed**	**High** (secondary)	High (primary)
Fasting urinary calcium	**Normal** (< 0.11 mg/dL glomerular filtrate)	**High**	High
Serum phosphate	Normal (low in phosphate leak)	Normal	Low
Sequence	Hyperabsorption → PTH suppressed → ↑ filtered load	Renal loss → PTH ↑ → calcitriol ↑	PTH → bone resorption + calcitriol → hypercalcaemia

Absorptive hypercalciuria

- **Type I**: persists on calcium restriction; **type II**: normalises with dietary restriction.
- **CYP24A1** loss-of-function (24-hydroxylase cannot inactivate calcitriol) → high calcitriol, idiopathic infantile hypercalcaemia; in adults hypercalcaemia, hypercalciuria, nephrocalcinosis, stones — worse with vitamin D supplements.
- **Renal phosphate leak**: phosphate wasting → hypophosphataemia → ↑ calcitriol → ↑ gut calcium absorption. Hereditary hypophosphataemic rickets with hypercalciuria (SLC34A1/SLC34A3, sodium–phosphate co-transporters NaPi-IIa/IIc).

Renal hypercalciuria — monogenic models

Disorder	Gene / protein	Features
Dent disease (X-linked)	Chloride channel-5 (ClC-5)	Hypercalciuria, **low-molecular-weight proteinuria**, nephrocalcinosis, stones, progressive renal failure
Familial hypomagnesaemia with hypercalciuria and nephrocalcinosis	Claudin-16 (paracellin-1), claudin-19	Loss of paracellular magnesium and calcium reabsorption in TAL
Bartter syndrome	Na^+-K^+-$2Cl^-$ co-transporter (NKCC2), ROMK potassium channel, ClC-Kb, barttin; type V = activating CaSR	Salt wasting, hypokalaemic alkalosis, hypercalciuria, nephrocalcinosis

> KEY POINT **Fasting urine + PTH separate gut from kidney**: renal leak → fasting urinary calcium high, PTH high, serum calcium normal; absorptive → fasting urine normal, PTH suppressed. The exact mirror images.

Resorptive and hypercalcaemic hypercalciuria

- **PHPT** causes ≈ 5% of nephrolithiasis; stones occur in < 5% of PHPT. Suspect when serum calcium > **10.1 mg/dL**; repeat with intact PTH; ionised calcium helps when total calcium is normal (normocalcaemic PHPT).
- **Thiazide challenge**: thiazide ↑ renal calcium reabsorption → worsens hypercalcaemia **without suppressing PTH** in PHPT — unmasks it. PTH also causes phosphaturia and mild hyperchloraemic acidosis.

Cause	Mechanism	PTH	Clue
PHPT	Autonomous PTH	High / inappropriately normal	Commonest hypercalcaemia in outpatients
Sarcoidosis (granulomas)	Macrophage 1α-hydroxylase → extrarenal calcitriol	Suppressed	Rapid correction with **corticosteroids** (cut gut absorption)

Cause	Mechanism	PTH	Clue
Malignancy	PTH-related protein (PTHrP), osteolysis → receptor activator of nuclear factor-κB ligand (RANKL)-driven osteoclasts	Suppressed	Commonest hypercalcaemia in inpatients (lung, breast)
Glucocorticoid excess	↑ Bone resorption	Raised	Stones common in Cushing syndrome

- Others: thyrotoxicosis, vitamin D toxicity.

8 · HYPEROXALURIA

- **Definition**: urinary oxalate > **40 mg/day**. Raises calcium oxalate saturation.
- Three types: primary, enteric, dietary (plus idiopathic).

Primary hyperoxaluria (PH)

- Autosomal recessive hepatic defects of glyoxylate metabolism: glyoxylate cannot become glycine, so it is oxidised to oxalate. Urinary oxalate ≥ **100 mg/day** → stones, marked nephrocalcinosis, renal injury → falling glomerular filtration rate (GFR) → **systemic oxalosis** (vicious circle).

Feature	primary hyperoxaluria type 1 (PH1)	PH2	PH3
Enzyme	Alanine-glyoxylate aminotransferase (AGT), pyridoxal phosphate (vitamin B6)-dependent; liver peroxisome	Glyoxylate reductase/hydroxypyruvate reductase (GRHPR)	4-Hydroxy-2-oxoglutarate aldolase (HOGA1); mitochondria
Urine markers	Oxalate + **glycolate**	Oxalate + **L-glyceric acid**	Oxalate; hypercalciuria
Course	**Most aggressive**; ESRD 2nd–3rd decade	Milder	Childhood stones; ESRD very rare
Transplant	**Combined liver–kidney**	Isolated kidney	Rarely needed

- **Pyridoxine responsiveness**: commonest AGT gene mutation Gly170Arg (with Pro11Leu polymorphism) mistargets AGT to mitochondria; pyridoxal phosphate (AGT cofactor) raises activity and restores peroxisomal targeting.
- **Why liver–kidney transplant in PH1**: the liver is the only organ that detoxifies glyoxylate; a kidney alone is re-injured by continuing hepatic oxalate overproduction.
- Genetic testing confirms type, allows prenatal diagnosis and family screening; **exclude PH in relatives before living-related donation**.
- †Lumasiran (RNA-interference against glycolate oxidase, reducing glyoxylate supply) is approved for PH1 → *Ch 99*.

Enteric hyperoxaluria — commonest acquired cause

ALGORITHM
Fat malabsorption → luminal fatty acids saponify calcium and magnesium → less calcium to bind oxalate → more free oxalate → **colonic hyperabsorption** (bile salts and fatty acids also ↑ colonic permeability)

- Causes: small-bowel resection or disease (Crohn disease), jejunoileal bypass, cystic fibrosis, malabsorptive bariatric surgery.
- **Requires an intact colon.** Accompanying risks of chronic diarrhoea: dehydration, hypokalaemia, hypocitraturia, hypomagnesiuria, low urine pH.
- Bariatric: Roux-en-Y gastric bypass and duodenal switch (malabsorptive) ↑ oxalate; banding and sleeve (restrictive) do not.

	Colon intact (short bowel)	Colon absent (ileostomy)
Oxalate	Hyperoxaluria	Normal

	Colon intact (short bowel)	Colon absent (ileostomy)
Typical stone	Calcium oxalate	Uric acid, ammonium acid urate (low volume, pH, sodium)

Dietary, microbiome, idiopathic

- Oxalate-rich foods: nuts, chocolate, brewed tea, spinach, potatoes, beets, rhubarb. **Severe calcium restriction is counter-productive** — less luminal binding → more oxalate absorbed. Vitamin C supplements ↑ urinary oxalate.
- Stone formers have less *O. formigenes* colonisation; antibiotics deplete it and raise stone risk.
- Mild idiopathic hyperoxaluria may matter as much as hypercalciuria (e.g. Arabian Peninsula, where hypercalciuria is rare).

9 · HYPERURICOSURIA, HYPOCITRATURIA AND HYPOMAGNESIURIA

Hyperuricosuric calcium nephrolithiasis

- **Definition**: urinary uric acid > **600 mg/day**; commonest cause dietary purine excess. Others: gout, myeloproliferative disease, hypoxanthine-guanine phosphoribosyltransferase (HGPRT) deficiency.
- Three ways urate promotes calcium oxalate: (1) colloidal urate **adsorbs inhibitors**; (2) **salting out** — dissolved urate lowers calcium oxalate solubility; (3) **heterogeneous nucleation/epitaxy** on urate crystals.
- Epidemiology conflicting (urinary uric acid not positively associated with incident stones in cohorts), but allopurinol cut recurrence in hyperuricosuric formers (see trials).

Hypocitraturia

- **Definition**: urinary citrate < **320 mg/day**.
- **Acid–base status is the main determinant**: acidosis → ↑ proximal citrate reabsorption and ↓ synthesis.

Cause	Mechanism
Distal RTA	Systemic acidosis; urine pH high
Chronic diarrhoea	Gut alkali loss
Animal protein excess	Acid load
Thiazides	Hypokalaemia → intracellular acidosis
Angiotensin-converting enzyme (ACE) inhibitors	Possibly intracellular acidosis

> **KEY POINT** Citrate excretion reads proximal-tubular **intracellular pH**: anything acidifying the cell (acidosis, hypokalaemia, acid diet) makes the tubule reclaim citrate — which is why alkali (potassium citrate) is the treatment → *Ch 99*.

Hypomagnesiuria and low urine pH

- Hypomagnesiuria: rare; favourable urine chemistry, but **two randomised trials of magnesium showed no clinical benefit**.
- **Low urine pH** (< 5.5): undissociated uric acid predominates → uric acid stones and calcium oxalate by heterogeneous nucleation. "Idiopathic low urine pH" = former "gouty diathesis". Chronic metabolic acidosis → low pH, hypercalciuria, hypocitraturia.

10 · RENAL TUBULAR ACIDOSIS

- RTA = hyperchloraemic (normal anion gap) metabolic acidosis from defective H^+ secretion or bicarbonate reabsorption with inappropriately high urine pH.
- Normal: proximal tubule reclaims bicarbonate; α-intercalated cells excrete H^+ (apical H^+-ATPase, basolateral anion exchanger 1 [AE1]) as titratable acid and ammonium (urine pH down to 4.5).

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4
Defect	Distal H^+ secretion	Proximal HCO_3^- reabsorption	Aldosterone deficiency/resistance, ↓ ammoniagenesis
Serum K^+	Low	Low	**High**
Urine pH	**Always > 6.0**; cannot go < 5.5	High, then < 5.5 at steady state	Low; can acidify
Urinary citrate	**Profoundly low**	Relatively normal	—
Stones	**Up to 70%** — calcium phosphate, nephrocalcinosis	Uncommon	Uncommon (uric acid if any)
Associations	Sjögren, lupus, obstruction, hypercalciuria; genetic	Fanconi syndrome	Diabetic nephropathy, interstitial disease

- **Type 1** matters most to urologists: commonest form; stones lead to the diagnosis in up to half. Calcium phosphate stones from **hypercalciuria** (acidosis → bone demineralisation), **profound hypocitraturia** (probably the most important factor) and **high urine pH**.
- **Incomplete distal RTA**: cannot lower urine pH < 5.5 after an acid (ammonium chloride) load, but serum electrolytes normal.
- Genetics: SLC4A1 (AE1; mostly autosomal dominant); ATP6V1B1, ATP6V0A4 (H^+-ATPase; autosomal recessive, severe, **sensorineural hearing loss**).
- **Type 2** is self-limiting once plasma bicarbonate falls; citrate spared → few stones.

> **KEY POINT** Stone risk in RTA follows citrate and pH: only distal RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria — the recipe for **calcium phosphate** stones.

11 · URIC ACID STONES

- Humans lack hepatic **uricase** (gene silenced in evolution), so uric acid is not converted to the 10–100× more soluble allantoin.
- Uric acid pKa **5.35**; sodium urate ≈ 20× more soluble than free acid. Solubility of undissociated uric acid ≈ 96–100 mg/L — easily exceeded below pH 6; at pH 6.5 > 1200 mg/L stays soluble.
  †Pure uric acid stones are radiolucent on plain kidney–ureter–bladder (KUB) radiograph but visible on non-contrast CT (NCCT).

Determinant	Causes	Importance
Low urine pH	Insulin resistance/obesity, animal protein, chronic diarrhoea, gout	**Single most important** — most formers have normal uric acid excretion
Low volume	Diarrhoea, hot climate/occupation	Concentrates uric acid
Hyperuricosuria	Purine diet, gout, myeloproliferative disease, uricosurics	Least often the driver

Why the urine is acid — insulin resistance

ALGORITHM
Insulin resistance / renal lipotoxicity (tubule burns fatty acids not glutamine; ↓ sodium–hydrogen exchanger 3 [NHE3]) → ↓ proximal ammoniagenesis → H^+ buffered by titratable acid instead of ammonium → **persistently low urine pH** → undissociated uric acid → stone

- Uric acid formers have low ammonium, high titratable acid and higher net acid excretion. Uric acid forms about a third of stones in type 2 diabetics; BMI ≥ 27 kg/m² predicts urine pH ≤ 5.5.

	Idiopathic uric acid nephrolithiasis	Hyperuricosuric calcium nephrolithiasis
Urinary uric acid	Normal	High
Urine pH	Low	Normal

	Idiopathic uric acid nephrolithiasis	Hyperuricosuric calcium nephrolithiasis
Stone	Uric acid	Calcium oxalate

> **KEY POINT** Uric acid stones are a disease of **urine pH, not uric acid** — a renal manifestation of metabolic syndrome. Because there is no inhibitor, raising urine pH above the pKa removes the main driver; †alkalinisation can dissolve existing uric acid stones → *Ch 99*.

12 · CYSTINE STONES

- Cystine and the other dibasic amino acids (†ornithine, lysine, arginine) are reabsorbed in the proximal tubule and jejunum by a heteromeric transporter. Mutation in either subunit → **cystinuria**, autosomal recessive.

Subunit	Gene	Chromosome	International Cystinuria Consortium type
rBAT (heavy)	SLC3A1	2p	Type A
$b^{0,+}$AT (light)	SLC7A9	19q	Type B
Both	—	2 and 19	Type AB

- No nutritional deficiency — amino acids are absorbed as oligopeptides.
- **Only cystine stones form** — the other dibasic amino acids are soluble. Incidence 1 in 1000–17,000; up to 10% of paediatric stones; males more severe; first stone ≈ 12 years.
- **Pure supersaturation disease — no inhibitor**. Solubility ≈ 300 mg/L at pH 5, 400 mg/L at pH 7, 1000 mg/L at pH 9 — meaningful only at markedly alkaline pH.
- Coexisting hypocitraturia (commonest), hyperuricosuria, hypercalciuria → mixed calcium–cystine stones.
- Cystine formers have **worse renal function** (about a quarter of adults have estimated GFR < 60 mL/min/1.73 m^2) — repeated surgery, cystine duct plugging. †Crystals are hexagonal; †cyanide–nitroprusside test screens urine for cystine.

13 · INFECTION (STRUVITE) STONES

- Magnesium ammonium phosphate hexahydrate (struvite) ± carbonate apatite. **Urease is absent from sterile urine** — infection with a urease-producing organism is a prerequisite.

ALGORITHM
Urea → (urease) $2NH_3 + CO_2 \rightarrow NH_3 + H_2O \rightarrow NH_4^+ + OH^-$ → urine pH 7.2–8.0 (urease keeps producing ammonia, so pH keeps rising) → CO_2 → carbonate; phosphate → PO_4^{3-} → with magnesium: **struvite**; with calcium: carbonate apatite

- Bacteria damage the glycosaminoglycan layer → adherence → **biofilm**; struvite crystallises within the exopolysaccharide matrix, **encasing bacteria inside the stone** — why fragments harbour infection and complete clearance matters. Stasis accelerates growth.
- Organisms: ***Proteus mirabilis*** most associated; *Staphylococcus aureus* and **Ureaplasma urealyticum** (easily missed) usually urease-positive; Klebsiella and Pseudomonas occasionally. *Escherichia coli* rarely produces urease yet is often cultured — **bladder and stone cultures are discordant**, so voided urine culture is unreliable.
- 5–15% of stones and falling; female : male ≈ 2 : 1. **Less than half of staghorn calculi are infection stones** — most are metabolic (calcium phosphate).
- At risk: elderly, diabetes, stasis (obstruction, urinary diversion, neurogenic bladder), spinal cord injury.
- †Struvite crystals: "coffin-lid" shape on urine microscopy.

14 · MISCELLANEOUS AND DRUG-RELATED STONES

	Xanthine	2,8-Dihydroxyadenine (DHA)
Enzyme deficient	Xanthine dehydrogenase (XDH)	Adenine phosphoribosyltransferase (APRT)
Radiology	Both radiolucent — mistaken for uric acid	
Treatment clue	—	Allopurinol or febuxostat (febuxostat more effective)

- **Ammonium acid urate** (≈ 1%; endemic paediatric bladder stones): radiolucent; chronic diarrhoea, inflammatory bowel disease, ileostomy, **laxative abuse**, recurrent urinary tract infection (UTI), obesity.

Route	Drug	Mechanism / stone
Direct	**Indinavir**	≈ 50% excreted unchanged, insoluble at physiological pH (more soluble < 5.5); stones 4–13%; †radiolucent even on CT
Direct	Triamterene; guaifenesin/ephedrine; silicate antacids (magnesium trisilicate)	Drug/metabolite stones
Indirect	Carbonic anhydrase inhibitors (acetazolamide, **topiramate**, zonisamide)	Mimic distal RTA: acidosis, alkaline urine, hypocitraturia → calcium phosphate
Indirect	Loop diuretics	Hypercalciuria → calcium oxalate (up to 64% of low-birth-weight infants on furosemide)
Indirect	Corticosteroids, vitamin D, thiazides, laxatives, cytotoxics	Hypercalciuria, hypocitraturia, ammonium urate, uric acid

15 · ANATOMIC PREDISPOSITION AND PREGNANCY

Anomaly	Stone frequency	Metabolic findings	Verdict
Ureteropelvic junction obstruction (UPJO)	≈ 20%; children ~70× normal risk	Common, same spectrum as other formers (hypercalciuria first); recurrence after repair, often contralateral	Metabolic > anatomic
Horseshoe kidney	≈ 36% adults	At least one abnormality in all tested; **hypocitraturia over-represented**; calcium stones ≈ 89%	Stasis + metabolic
Calyceal diverticulum	Up to 40%	Abnormalities in half; UTI up to 40%	Metabolic + stasis
Medullary sponge kidney (MSK)	Frequent, with nephrocalcinosis	Hypercalciuria, hypocitraturia; distal RTA on acid loading; H^+-ATPase mutations	Metabolic ± ductal stasis

> **KEY POINT** Anatomy sets the stage but metabolism writes the stone: patients with anatomical stones still need full **metabolic evaluation**, and surgical correction alone (e.g. pyeloplasty) does not prevent recurrence — treating the metabolic defect substantially cuts it → *Ch 99*.

Pregnancy

- Symptomatic stones 1 in 204–3000 pregnancies; mostly 2nd–3rd trimester.
- **Physiological hydronephrosis** in up to 90%, **right > left**, persists 4–6 weeks post partum.
- Promoters: GFR ↑ 30–50% → ↑ filtered calcium, uric acid; placental calcitriol → absorptive hypercalciuria. Inhibitors: ↑ citrate, magnesium, glycoproteins. **Net risk unchanged**, but stones skew to **calcium phosphate**.

16 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Borghi water trial	Randomised controlled trial (RCT), high fluid intake vs no advice, first-time idiopathic calcium formers	Recurrence 12% vs 27%	Urine volume is the one proven universal modifier → *Ch 99*
Ettinger allopurinol trial	RCT, allopurinol vs placebo, hyperuricosuric normocalciuric calcium oxalate formers	> 2-fold fewer recurrences	Allopurinol for hyperuricosuric calcium oxalate stones → *Ch 99*

17 · MCQ ONE-LINERS

1. Commonest stone: calcium oxalate (≈ 60%); calcium stones ≈ 80% overall.
2. Most urinary salts sit in the metastable zone — where inhibitors and therapy act.
3. Urine pH changes affect phosphate, uric acid and cystine — not oxalate.
4. Randall plaque: calcium apatite, originating in thin-limb basement membrane.
5. Ductal plugs suggest a secondary cause (enteric hyperoxaluria, distal RTA, cystinuria).
6. No known urinary inhibitor of uric acid or cystine.
7. Calcitriol: most potent stimulator of gut calcium absorption; PTH has no direct gut effect.
8. Hypercalciuria: > 200 mg/day on 400 mg calcium diet, or > 4 mg/kg/day.
9. Absorptive: normal fasting urinary calcium, suppressed PTH; renal: high fasting calcium, high PTH.
0. Thiazide challenge unmasks normocalcaemic PHPT.
1. Sarcoid hypercalciuria: extrarenal calcitriol, responds to corticosteroids.
2. Dent disease: ClC-5, low-molecular-weight proteinuria, X-linked.
13. Hyperoxaluria > 40 mg/day; primary hyperoxaluria ≥ 100 mg/day.
14. PH1: AGT deficiency, glycolate in urine; liver–kidney transplant.
15. Enteric hyperoxaluria needs an intact colon; ileostomy gives uric acid stones instead.
16. Hyperuricosuria > 600 mg/day; hypocitraturia < 320 mg/day.
17. Distal RTA: urine pH > 6, hypocitraturia, calcium phosphate stones, up to 70%.
18. Recessive distal RTA (H^+-ATPase B1) has sensorineural deafness.
19. Uric acid pKa 5.35; low urine pH is the dominant cause of uric acid stones.
20. Cystinuria: SLC3A1 (chromosome 2, type A), SLC7A9 (chromosome 19, type B).
21. Cystine solubility rises meaningfully only at alkaline pH (≈ 1000 mg/L at pH 9).
22. Struvite needs urease; Proteus mirabilis commonest; E. coli rarely urease-positive.
23. Most staghorn calculi are metabolic (calcium phosphate), not struvite.
24. Topiramate and acetazolamide cause calcium phosphate stones via distal RTA-like state.

96 Acute Care of Urinary Lithiasis

Reading time ≈ 13 min (slow pace)

1 · PRESENTATION

- **Renal colic**: unilateral severe flank pain radiating to lower abdomen/genitalia; not positional — patient writhes; waxes and wanes. Dull pressure → suspect chronic obstruction.
- Stone at the ureterovesical junction (UVJ): urethral-tip pain, frequency, urgency. Sudden relief → stone in bladder.
- Haematuria ~**90%**; absent in >10%.
- **Fever/chills → infected obstructed system** → urgent intervention.

2 · DIAGNOSIS

Laboratory

- Serum chemistry, urinalysis, complete blood count (CBC) with differential, **pregnancy test** in every woman of reproductive age.
- Raised white blood cells (WBCs) may be stress, not urinary tract infection (UTI); urine >50 white blood cells (WBC)/high-power field or nitrites suggest UTI.

Imaging

Modality	Role	Limitation
Non-contrast computed tomography (NCCT)	Gold standard (sensitivity 97%, specificity 95%); size, site, skin-to-stone distance, secondary signs, alternative diagnoses	Radiation (8–16 mSv); misses matrix and indinavir stones
Low-dose NCCT (≤**3 mSv**)	Equal for stones ≥3 mm	Weak <3 mm, obesity
Ultrasound (US)	First-line in pregnancy, children, known recurrent formers	Hydronephrosis as surrogate; up to 40% still need computed tomography (CT)
Plain kidney–ureter–bladder X-ray (KUB)	**Avoid** for acute diagnosis	Lowest accuracy despite radiopaque stones
Magnetic resonance imaging (MRI)	Second-line in pregnancy/children	Poor stone detection

- STONE score (five-criterion clinical score) failed validation as a substitute for NCCT.

> **KEY POINT** Unknown diagnosis → NCCT answers everything at once. Known stone → US showing hydronephrosis suffices.

3 · PAIN MANAGEMENT

- **Mechanism**: obstruction → pelvic/capsular stretch → prostaglandins → diuresis, vasodilation, rising intrarenal pressure + ureteral smooth-muscle spasm. Shared innervation → referred pain to gut, groin, testis/labium.
- **Non-steroidal anti-inflammatory drugs (NSAIDs)** block cyclo-oxygenase (COX) → treat the cause; equal to opioids with less rescue analgesia, vomiting and respiratory depression.

Agent	Place	Note
Ketorolac 30 mg (NSAID)	**First line** (European Association of Urology [EAU], National Institute for Health and Care Excellence [NICE])	Avoid in renal failure, gastrointestinal bleed, pregnancy
Intravenous paracetamol	Second line	NSAID contraindicated/insufficient

Agent	Place	Note
Opioids	Intractable pain	**Continue the NSAID** — morphine + ketorolac beats either
Antispasmodics; forced fluids/diuretics	**Ineffective**	Fluids only for vomiting/dehydration

- Opioid dependency rises with each day prescribed beyond day 3 → prescribe fewest days.

4 · MEDICAL EXPULSIVE THERAPY (MET)

- α-Blockers reduce spasm and peristaltic frequency → larger urine bolus and higher pressure behind the stone. NSAIDs alone do not aid passage.
- American Urological Association (AUA)/EAU: offer MET for **distal ureteral stones <10 mm** not needing immediate intervention; tamsulosin †(0.4 mg daily) most studied. Silodosin expels better, but retrograde ejaculation ~2×.
- Benefit confined to **distal stones 5–10 mm**; none ≤5 mm or proximal.
- Trial of passage ± MET for **4–6 weeks**, then intervene.

Width	Passage	Location	Passage
0–3 mm	98%	UVJ	92%
5 mm	65%	Distal	75%
6 mm	33%	Mid	60%
>6.5 mm	9%	Proximal	48%

Measure **transverse width**, not length.

5 · URGENT INTERVENTION

- **AUA indications**: obstruction + infection; bilateral obstruction; solitary kidney; immunosuppression; impending renal deterioration; refractory pain; intractable vomiting.
- Uninfected obstruction harms function only if prolonged (>~2 weeks).
- Decompression: **retrograde stent or percutaneous nephrostomy (PCN)** — equal for sepsis resolution; stent → more dysuria/haematuria; PCN → longer stay.
- **Stent failure predictors**: pericalcular ureteric thickness (P-CUT) ≥**7.5 mm** (strongest); transverse stone diameter ≥9.5 mm; multiple/large stones → prefer PCN.
- P-CUT = ureteric transverse diameter at the stone (CT) − stone width.
- Acute shock wave lithotripsy (SWL, proximal stones, ideally <24 h) or acute ureteroscopy (URS; distal, <10 mm) **only without infection** → *Ch 97*.

ALGORITHM
Colic → analgesia + NCCT → infection/AUA criterion? Yes → drainage only (stent/PCN) → definitive later. No → stone <10 mm → trial ± MET 4–6 weeks → fails → URS/SWL

6 · SPECIAL POPULATIONS

Pregnancy

- Symptomatic in <1%; risk of preterm birth, pre-eclampsia.
- Imaging: **US → MRI → low-dose NCCT** (American College of Obstetricians and Gynecologists [ACOG]). Physiological hydronephrosis ~90%, rarely below iliac vessels; resistive index difference ≥**0.06** vs contralateral = obstruction. Fetal risk not significant <50 mSv.
- Conservative first; ~30% need intervention. Stent/PCN encrust → change every **4–6 weeks**; early gestation favours PCN (each stent exchange needs anaesthesia). Preterm delivery highest with PCN.

- **URS is safe in any trimester** (ACOG 2019) and avoids repeated exchanges.

Population	Pitfall	Key point
Elderly/frail	Atypical/no pain; first stone >60 is unusual	Always CT — exclude aortic and cardiac mimics; NSAIDs often barred; early drainage
Transplant kidney	Denervated → **painless** obstruction	Fever/renal failure; endoscopic, else guided percutaneous access

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SUSPEND	Randomised controlled trial, tamsulosin vs placebo, any ureteral stone, n = 1136	Intervention at 4 weeks 19% vs 20%	Unstratified; does not abolish MET for distal 5–10 mm stones

8 · MCQ ONE-LINERS

1. Gold standard for acute colic: NCCT.
2. NCCT misses indinavir and matrix stones.
3. KUB unacceptable for acute diagnosis.
4. First-line imaging in pregnancy/children: US.
5. Haematuria absent in >10% of stones.
6. First-line analgesia: NSAID (ketorolac 30 mg).
7. Antispasmodics and forced fluids: no benefit.
8. MET benefit: distal stones 5–10 mm.
9. Silodosin: better expulsion, more retrograde ejaculation.
10. Trial of passage limit: 4–6 weeks.
11. Passage predicted by transverse width.
12. Stone >6.5 mm wide: ~9% pass.
13. Infection + obstruction: drain only.
14. Strongest stent-failure predictor: P-CUT ≥7.5 mm.
15. Pregnancy imaging order: US → MRI → low-dose CT.
16. Pregnancy hydronephrosis distal to iliac vessels suggests stone.
17. Resistive index difference ≥0.06 indicates obstruction.
18. URS permissible in any trimester.
19. First stone after 60: exclude aortic and cardiac causes.

97 Surgical Management of Ureteral Calculi

Reading time ≈ 29 min (slow pace)

1 · BACKGROUND AND DIAGNOSIS

- Pain is from capsular distension; the dangers are **renal damage and sepsis**.
- Pain mimics appendicitis or aortic aneurysm → imaging decisive; absent hydronephrosis does not exclude a stone.

Modality	Strength	Limitation
Non-contrast computed tomography (NCCT)	Gold standard; low-dose equally effective; replaced intravenous pyelography	Radiation
Ultrasound (US)	No radiation	Misses mid-ureter (between ureteropelvic junction [UPJ] and ureterovesical junction [UVJ])
US + kidney–ureter–bladder radiograph (KUB)	Good follow-up once NCCT has localised the stone	Radiolucent stones invisible

- Medical expulsive therapy (MET) with α-blockers remains controversial after SUSPEND → *Ch 96*. Passage becomes less likely with time → intervene after **4–6 weeks**.
- Ladder: shock wave lithotripsy (SWL) → ureteroscopy (URS) → percutaneous nephrolithotomy (PCNL) with antegrade URS.

2 · INDICATIONS AND PREPARATION

Tier	Trigger
Elective	Failed trial of passage
Urgent	Unrelenting pain; obstruction-related renal dysfunction
Extremely urgent	Obstructive pyelonephritis/sepsis → **drain only** (nephrostomy or stent), never manipulate the stone

- Work-up: blood count, creatinine, **urine culture**; exclude coagulopathy and untreated urinary tract infection (UTI). Preoperative KUB confirms radio-opacity for SWL and allows KUB + US follow-up instead of repeat NCCT.

Modality	Antiplatelet/anticoagulant	Bridging
PCNL	Stop if feasible	Short-acting heparin if high thrombotic risk
SWL	Stop if feasible	Not recommended — if needed, choose another modality
Laser URS	**Need not stop**	None; more haematuria, worse stent symptoms

> **KEY POINT** Why only URS tolerates anticoagulation: SWL and PCNL injure parenchyma where bleeding is unseen and uncontrollable; laser URS works intraluminally under vision.

3 · CHOOSING TREATMENT

- Factors: surgeon skill; patient (comorbidity, anatomy, skin-to-stone distance [SSD], preference); stone (site, size, density, impaction, infection); equipment.
- Both SWL and URS treat stones anywhere in the ureter. Upper ureter ≤2 cm: similar stone-free rate (SFR) at 3 months, more SWL retreatment. Distal: similar SFR, SWL more second sessions but fewer complications.
- **Impaction** (thick ureteral wall around the stone on NCCT/US) defeats SWL → endoscopic treatment.

- Large, hard proximal stones → PCNL + antegrade URS; if unfit for PCNL, staged retrograde URS.

Modality	Size guidance
SWL	Historical limit 2 cm; SFR falls above **10–15 mm**
URS	No definite limit
PCNL ± antegrade URS	Consider >15–20 mm, large hard proximal stones

> **KEY POINT** Constant trade-off: SWL = fewer complications, more sessions; URS = one session, slightly more morbidity.

4 · URETEROSCOPES

Instrument details are covered with renal stones → *Ch 98*; ureteral essentials:

	Semirigid	Fibreoptic flexible	Digital flexible	Single-use digital
Image	Stable unless bent	Degrades as fibres break	No degradation	New each case
Calibre	Tapered; dilates orifice	Smallest tip	Larger, flatter tip (~8.5 Fr)	7.7 French (Fr) tip
Cost	Cheap	Repair ~every **15** cases		Suits low volume

- **Semirigid**: damaged by torque at the pelvic brim and laser near the tip; for distal/mid ureter.
- **Flexible**: 270° deflection, 3.6 Fr channel; suits proximal stones — follows a stone that migrates into the kidney.
- Digital (complementary metal-oxide semiconductor [CMOS] "chip-on-tip") is **not more durable**; keep a fibreoptic scope for a narrow UPJ/ureter. Scope-sparing: 200 µm fibres, access sheath, no deflection inside it.
- Raised intrarenal pressure (IRP) → absorption, fever, sepsis; integrated scope sensors now monitor it.

5 · CONSENT AND ACCESS

- Consent: access failure, incomplete clearance, further procedures, perforation, avulsion, sepsis, stricture, loss of renal function/nephrectomy.
- First step: identify the orifice, place a **safety wire** (Segura's dictum: only a safety wire separates endourology from a motorcycle accident). "Wireless" URS is safe in uncomplicated cases, but complexity is unpredictable → most advise a wire.
- **Difficult ureter**: manoeuvre now, or stent for passive dilation and return later (extra anaesthetic, but among the safest). Second-look after pre-stenting: **2–4 weeks** (≥14 days for SFR; ≤4 weeks for lower sepsis).

Guidewires

	Hydrophilic	Polytetrafluoroethylene (PTFE)-coated	Hybrid
Build	Nitinol, lubricious when wet	Steel or nitinol core	Floppy hydrophilic tip, stiff PTFE shaft
Best for	**Bypassing** impacted stones	Safety/working wire; stays put	Difficult anatomy + instrument passage
Weakness	**Easily displaced**	Damages flexible-scope channel on backloading	—

- Uncertain wire path → open-ended catheter + retrograde pyelogram (RPG).

Wire will not pass the stone

ALGORITHM
Open-ended catheter to 1–2 mm short of stone → RPG outlines obstruction → contrast hydrodissection + hydrophilic floppy-tip wire (catheter directs force) → advance catheter past stone → exchange for stiff wire (straightens tortuous ureter)

- Catheter abutting the stone → wire exits at 90° → **perforation**. Alternative: lithotripsy first, then place the safety wire under vision.
- ~1 in 10 ureters cannot be accessed safely → **double-J stent and stage** rather than risk injury.

Scope will not enter the orifice

- Persistence risks mucosal flap, false passage, perforation. Options: **railroading** between two wires (semirigid); dual-lumen catheter; rotating the flexible scope (offset channel); sheath obturator as dilator.

Dilation method	Best for
Indwelling stent (passive)	Whole ureter narrow — gentlest, but commits to second procedure
Serial/coaxial (8–10 Fr) dilators	Long narrow segments
Balloon (4–10 cm)	Focal stenosis — inflate until waist resolves, hold 1–2 min
Semirigid scope	Distal ureter

- **Pre-stenting** eases access but is **not needed** for ureteral stones (no SFR gain in meta-analysis and global database; benefit only renal/flexible URS).
- **α-Blockers ~1 week pre-URS** (α1 receptors abundant distally): less dilation, better access, higher SFR, less stent pain — use judiciously (distal stones, large scope or sheath).

Ureteral access sheath (UAS)

- Helps with large burden (repeated passes), dusting (outflow for vision) and high laser power (heat removal); lowers IRP and temperature. Commonest 12/14 Fr.
- Placed over a stiff working wire under fluoroscopy, safety wire outside. Hard to stabilise low → reserve for stones **above the iliac vessels**, tip just below the stone.
- Larger sheaths (14/16 Fr) reduce ureteral blood flow → stricture risk, clinically rare.

6 · LITHOTRIPSY

Basketing

- Keep stone **5–10 mm from scope tip** on withdrawal; **never basket blind**. Align long axis with ureter.
- **Entrapped stone**: push into wider segment → disassemble basket, shaft as wire → scope alongside → laser stone (or nitinol basket) → else cut basket distal to handle.
- When in doubt, **fragment before extracting** (prevents perforation and avulsion).

Energy sources

Source	Key points
Electrohydraulic	Omnidirectional pressure wave; high ureteral trauma; abandoned
Pneumatic	Jackhammer probe, ~12 Hz; **rigid/semirigid only**; best below pelvic brim; retropulsion; pressing on fragments can perforate
Holmium:yttrium-aluminium-garnet (Ho:YAG)	Short pulse (SP) fragments; long-pulse/pulse-modulated reduce retropulsion; dusting = low energy, high frequency
Thulium fibre laser (TFL)	Low peak power, long pulse → fine particles; minimal retropulsion; >40 Hz mostly misses stone

- **Antiretropulsion devices** (Stone Cone, Accordion, NTrap) behind the stone improve SFR with semirigid URS, where migration may end the case.

Setting	Ho:YAG	TFL
Fragmentation	0.6–1.0 J × 6–10 Hz	Start 0.6 J × 6 Hz SP (3.6 W)
Dusting	0.2–0.3 J × 40–80 Hz	0.2 J × 40 Hz (8 W)

- **Impacted stone (TFL)**: target the centre first so sides fall into the lumen away from the wall; raise energy in small steps; poor view → lower frequency, raise energy. Paint, do not drill (TFL drilling melts stone).
- Randomised trials: TFL at low settings faster than Ho:YAG with equal ureteral SFR; at high Ho:YAG-style settings equivalent → **settings, not only the laser, decide outcome**.

Thermal safety

- Tissue damage above **43 °C**; ureter holds a small volume that heats fast. **Heat tracks power, not laser type.**
- Cap power at **≤10 W in ureter**, ≤25 W in kidney; sheath for outflow; continuous room-temperature or cold irrigant. Stopping irrigation to limit retropulsion → thermal injury and stricture.

7 · AFTER URS: ANTIBIOTICS AND STENTS

- **Single perioperative intravenous antibiotic dose** = postoperative course; extend only if urosepsis occurred in this stone episode.
- Stents cause symptoms in ~4/5. Stentless uncomplicated URS is safe (randomised trial; Cochrane: same pain, readmission, reintervention). Yet >half stent routinely.
- Removal on string = cystoscopic removal; preferred by patients.

Stent required	Decide at end of case
Perforation; stricture needing dilation; tight ureter; infection; planned second stage	After sheath use — inspect ureter on withdrawal; pre-stented + sheath + normal ureter → no stent

8 · SWL FOR URETERAL CALCULI

- Only truly non-invasive option; sedation or local anaesthesia, no stent, no general anaesthesia (GA). Guideline option in all ureteral locations.
- Acute obstructing stone: SWL **within 48 h** beats delayed SWL.

Absolute contraindications	Relative
Pregnancy; uncontrolled bleeding disorder; anticoagulation that cannot stop (treatment + 48 h); distal obstruction; untreated UTI; untreated/severe hypertension	Skeletal deformity or obesity preventing focusing; adjacent arterial aneurysm

- **Predictors**: size most important (largest linear dimension); gap favours URS ≥10 mm. Chronic impaction possibly strongest — NCCT "ring" sign (soft tissue around stone) with severe hydronephrosis, or stone static on serial imaging. Cystine, brushite, calcium oxalate monohydrate fragment poorly. Hounsfield units (HU) >1000 and SSD >10 cm proven for renal, mixed for ureteral stones.
- Preparation: **no routine pre-stenting** (no SFR gain; worse lower urinary tract symptoms [LUTS]). Hold aspirin, non-steroidal anti-inflammatory drugs (NSAIDs) and anticoagulants for treatment + 48 h, especially upper ureter (organs in blast path). Culture; antibiotics only with stents/tubes, struvite, recent UTI, reconstructed tract.
- Technique (detail → *Ch 98*): GA/sedation (10–15 mm movement impairs fragmentation); upper ureter supine, mid-ureter over sacrum **prone**, below sacroiliac joint supine; bubble-free coupling; **60–90 shocks/min** beats 120; ramp energy (protective vasoconstriction).

- Failure: repeat SWL vs salvage URS by fragmentation; little gain beyond 2 sessions; mid/distal retreat after 1–2 days. Salvage URS after failed ureteral SWL is **not harder**.
- Adjuncts: α-blockers raise SFR; **forced diuresis useless**.

	SWL	**URS**
SFR	Lower after one session (≥10 mm, impacted)	Higher early; equal at 3 months
Retreatment	More	Less; more stents and pain
Complications	Fewer, including UTI	Perforation

9 · SPECIAL SITUATIONS

- **Pregnancy**: conservative first (if no infection, solitary kidney, anuria, fetal distress); **SWL contraindicated**; URS safe under any anaesthesia with shielded limited fluoroscopy. Stents encrust faster → change **every 4 weeks**; retrograde stent before nephrostomy.
- **Urinary diversion/reimplant**: retrograde access often impossible → SWL (often prone) or **antegrade percutaneous flexible URS** (dilated ureter, dust washes down). Transplant: antegrade and retrograde both safe.
- **Congenital anomalies** (ectopia, duplication): retrograde URS usually feasible; escalate SWL → PCNL → laparoscopic ureterolithotomy.

10 · COMPLICATIONS

Complication	**Incidence**	**Management**
Perforation	0–4%	Stop lithotripsy → stent ~4 weeks → saline irrigant, drain urinoma, antibiotics → imaging after removal
Extruded stone	—	Leave it; stent; inform patient (may mimic new stone)
Submucosal stone	Chronic obstruction	Laser free + stent
Avulsion	0.06–0.5%	**Nephrostomy + delayed repair**; no immediate repair, avoid nephrectomy
Stricture	3–6%	By length and site; image every patient (silent obstruction). Risks: impaction, perforation, radiation
Sepsis	~5%	Broad-spectrum antibiotics, support; culture-directed

- Perforation causes: scope, forced sheath, unseen wire slippage, laser on wall, lithotrite/cautery (full thickness), baskets, poor vision. Difficult and not progressing → **stop, stent, stage**. Further lithotripsy risks fragments in the wall → granuloma → stricture.
- Avulsion: forceful extraction of large/impacted stone; **proximal third** (least muscle). Prevent: no blind basketing, fragment large stones, release or disassemble an oversized engaged basket.
- Sepsis risks: positive culture, pre-stent, age, diabetes, ischaemic heart disease, long surgery → treat culture preoperatively, keep IRP low.

> **KEY POINT** Nearly every serious URS complication is force without vision — blind basketing, forced sheaths, persistence in a tight ureter. Remedy: see, fragment, and when in doubt stop, stent and stage.

11 · FUTURE

- Robotic URS (ergonomics, less radiation); **burst wave lithotripsy** — handheld low-amplitude ultrasound bursts shatter stones in clinic, little anaesthesia.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Stentless URS	Randomised controlled trial (RCT), stent vs none after uncomplicated URS	Safe without stent	Stent only for named indications
TFL vs Ho:YAG	RCT, low TFL settings	TFL faster; equal ureteral SFR	Low settings optimal for TFL

13 · MCQ ONE-LINERS

1. US misses mid-ureteral stones.
2. Obstructed infected kidney: drain only.
3. Laser URS: anticoagulation may continue.
4. SWL SFR falls above 10–15 mm.
5. Pneumatic lithotripsy: never via flexible scope.
6. Hydrophilic wire: best to bypass, easily displaced.
7. PTFE wire damages flexible-scope channel on backloading.
8. Sheath tip placed above iliac vessels.
9. Pre-stenting unnecessary for ureteral stones.
0. URS after pre-stenting: 2–4 weeks.
1. Thermal injury threshold: 43 °C.
2. Ureteral laser power cap: 10 W.
13. Laser heat depends on power, not laser type.
14. TFL drilling melts stone — paint instead.
15. Single perioperative antibiotic dose suffices.
16. SWL rate 60–90 shocks/min beats 120.
17. Mid-ureteral stone over sacrum: SWL prone.
18. Strongest SWL failure predictor: impaction (ring sign).
19. Forced diuresis after SWL: no benefit.
20. Pregnancy: SWL absolutely contraindicated; stents changed 4-weekly.
21. Avulsion: commonest proximal third; nephrostomy, delayed repair.
22. Post-URS stricture: 3–6%; image every patient.

98 Surgical Management of Upper Urinary Tract Calculi

Reading time ≈ 34 min (slow pace)

1 · CHOOSING THE APPROACH

- Framework: 2016 American Urological Association (AUA)/Endourological Society guideline, built on index patients and shared decision making. Pathogenesis → *Ch 95*; ureteral stones → *Ch 97*.
- **Governing principle**: the more invasive the procedure, the more stone it clears — set against operative risk, patient goals and risk tolerance.

Domain	Factors
Stone	Size, number, location, composition
Renal anatomy	Obstruction/stasis, hydronephrosis, ureteropelvic junction obstruction (UPJO), calyceal diverticulum, horseshoe/ectopic kidney, lower pole
Patient	Infection, obesity, habitus deformity, coagulopathy, age, hypertension, renal failure/transplant, solitary kidney, diversion, pregnancy

KEY POINT Only the stone domain is changed by choosing a modality; anatomy and patient factors decide which modality is available. Obesity, coagulopathy and pregnancy each exclude *a* modality, not surgery.

Asymptomatic renal stones

- Truly asymptomatic = no symptoms and no sequelae (including no recurrent urinary tract infection [UTI]). Natural history not benign: **~50%** have a stone event within **5 years**.
- Predictors of an event: number, prior stone history, size, renal pelvic location, male sex, younger age. **Size predicts intervention, not symptoms** (risk rises >5 mm and >10 mm).
- **Recurrent UTI + non-obstructing stone**: remove regardless of composition; **culture the stone** to confirm the nidus and tailor antibiotics.
- Residual fragments **>4 mm** after ureteroscopy (URS) → treat (more growth, events, reintervention).
- **Treat small asymptomatic contralateral stones** during surgery on the symptomatic side (see trials).
- Declined treatment → medical prevention → *Ch 99* + serial imaging.

Preoperative evaluation

- Imaging: ultrasound, plain kidney–ureter–bladder radiograph (KUB), low-dose computed tomography (CT); intravenous urography obsolete.
- AUA: non-contrast CT (NCCT) **should** precede percutaneous nephrolithotomy (PCNL); **may** be used to choose between shock wave lithotripsy (SWL) and URS.
- SWL predictors on CT: stone-free rate (SFR) best with density **<900 HU** (Hounsfield units) and skin-to-stone distance (SSD) **<10 cm**.
- Functional imaging if ipsilateral function loss suspected; contrast CT for complex anatomy.
- All: urinalysis/culture (conventionally within 1 month); add blood count and electrolytes for comorbidity or bleeding risk.

Modality selection

Scenario (AUA)	Offer	Caveat
<2 cm, non-lower pole, symptomatic	SWL or URS	SWL fewer complications; URS higher SFR
Total burden >2 cm	**PCNL**; staged URS an option	SWL not first-line

Scenario (AUA)	Offer	Caveat
Lower pole >1 cm	PCNL (highest SFR) or URS	SWL not first-line — fragments clear poorly
Lower pole <1 cm	SWL or URS	—
Staghorn	PCNL	—
Cannot stop antiplatelet/anticoagulant	**URS**	Avoid SWL and PCNL

- European Association of Urology (EAU): >20 mm → PCNL, then URS/SWL; **10–20 mm → SWL or endourology in explicit equipoise**; <10 mm → SWL or URS, then PCNL (lower pole 10–20 mm handled separately).
- Staged URS avoids PCNL bleeding at the cost of more total operating time.

Stone morphometry scores (predict PCNL outcome)

Score	Inputs	Predicts
Guy's	Location, number, anatomy	SFR
STONE	Size, tract length, obstruction, number, density (CT)	SFR, blood loss, operative time, stay
CROES nomogram	Size, location, number, prior treatment, staghorn, centre volume	SFR (KUB only); only multicentre-derived
S-ReSC	Calyceal distribution	SFR

None reliably predicts complications; no generalisable score exists. The **Triple D score** predicts **SWL**, not PCNL MCQ.

2 · SHOCK WAVE LITHOTRIPSY

Physics

- First human SWL 1980 (Dornier); the electrohydraulic HM3 remains the efficacy benchmark MCQ.
- Shock wave: peak positive **30–100 MPa**, then negative tail ≈ −10 MPa; generated at F1, focused on the stone at **F2**. A focal zone **wider than the stone** breaks better and tolerates respiratory movement.

Source	Strength	Weakness
Electrohydraulic (spark gap)	Very effective	Electrode wear; shock-to-shock variability; focus drifts up to 1 cm
Piezoelectric	Long life, stable focus, anaesthesia-free	Lowest energy, smallest focal volume
Electromagnetic (modern machines)	Consistent up to 10^6 shocks; high F2 energy with low skin energy	—

Mechanisms of comminution

1. **Tear and shear**: internal elastic waves reflect and superimpose; lost once fragments are smaller than the pulse.
2. **Spallation**: wave inverts at the distal stone–fluid interface → tension breaks the distal end.
3. **Quasistatic squeezing**: wave faster in stone than fluid → circumferential "hoop" compression.
4. **Cavitation**: negative phase forms bubbles whose collapse erodes the surface; but excess bubbles shield the stone.
5. **Dynamic squeezing**: 1–3 combined with a wide focus — the most powerful force in initial breakage.

Selection and contraindications

- Divert **infection stones** and multi-site stones to a more efficient modality.

- **Absolute contraindications**: pregnancy; uncorrected coagulopathy or current antiplatelet/anticoagulant use; untreated UTI; severe untreated hypertension; aortic/renal artery aneurysm (aorta >**4 cm**); obstruction distal to the stone; inability to target (skeletal deformity).

Technique — maximising success and safety

- **Coupling** is the key technical factor: gel straight from the jug as a large mound, never spread. Air pockets reduce fragmentation by up to ~40%.
- **Rate**: **60–90 shocks/min** beats 100–120 — fast rates build a cavitation cloud that shields the stone and injures tissue.
- **Ramping** voltage stepwise (low-energy pretreatment over ≥4 min) → renal vasoconstriction protects the kidney and improves breakage.
- Preparation: check contraindications and bleeding drugs → confirm stone visible on targeting (KUB/ultrasound) → urinalysis with reflex culture → treat bacteriuria. No prophylactic antibiotic if urinalysis negative and no risk factors (AUA, EAU).

Complications

- **Arrhythmia**: common; pause, or gate shocks to the cardiac refractory period.
- **Renal haematoma**: minority, mostly mild; rarely embolisation. Risks: age, poorly controlled hypertension, obesity, diabetes, thrombocytopenia, narrow-focus machines, high energy, more shocks.
- **Steinstrasse**: rises several-fold with burden >2 cm^2. **Do not stent routinely at SWL** — no gain in passage, only stent morbidity. †Obstructed or infected steinstrasse → nephrostomy or URS.
- Alpha-blocker medical expulsive therapy (MET) speeds clearance and reduces pain; percussion–diuresis–inversion also helps.
- **Refuted**: diabetes, hypertension, long-term renal decline, infertility MCQ.

Burst wave lithotripsy (BWL) and ultrasonic propulsion

- Investigational, ultrasound-guided, awake. BWL uses narrowband bursts at **4–8 MPa**: same total energy as SWL at lower pressure over longer time → no anaesthesia, less cavitation, uniform small fragments.
- Propulsion moves fragments away from the transducer to aid passage.

3 · URETEROSCOPY (RETROGRADE INTRARENAL SURGERY, RIRS)

- Landmarks MCQ: Bozzini's Lichtleiter 1806 (first endoscope); Takayasu 1974 first ureteral access sheath (UAS).

Preoperative

- General anaesthesia: laryngeal mask usually fine, but intubation allows breath control → less renal excursion, more efficient lasering.
- Urine test before elective URS; single-dose antibiotic per antibiogram; pre-treat bacteriuria.
- **Antithrombotics**: URS is the modality of choice for those who must continue; anticoagulation appears safe, antiplatelets increase admissions. Hold non-indicated over-the-counter agents.

Instruments and principles

- **Safety wire**: recommended by AUA/EAU (EAU calls it controversial) — raises scope insertion forces; a multicentre study found no outcome difference. Never force a wire past an impacted stone (submucosal passage).
- **UAS**: lowers intrarenal pressure (larger sheath, lower pressure) and eases repeated passes for basketing — but meta-analyses show **no clear fall in infection**. Causes ureteral injury in nearly half (severe ~1 in 8), less if **pre-stented**; de novo stricture rare.
- Dusting vs basketing: equivalent SFR; basketing longer.
- **Irrigation**: working channels 3.6–4.6 Fr need pressure; hand pumps linked to more fever and emergency visits than gravity. Real-time intrarenal pressure sensors are promising.

- **Anti-retropulsion** devices (Stone Cone, NTrap, BackStop) reduce proximal migration.
- **Stent omission** (AUA) when all four: normal anatomy, normal renal function, no ureteral injury, no planned second stage — routine stents add symptoms without benefit.

Success

"Clinically insignificant fragments" (>5 mm cut-off, Lingeman) is flawed — smaller fragments cause reintervention in up to a fifth. Report zero-fragment rate plus >2 mm and >4 mm residual rates; SFR depends chiefly on imaging (CT strictest).

Complications

Complication	Rate	Risk factors / management
Sepsis	~5%	Pre-stent, positive culture, ischaemic heart disease, age, long surgery, diabetes → treat urine first
Subcapsular/perirenal haematoma	<1%	Hydronephrosis, thin cortex, long surgery, hypertension, UTI; rare but potentially devastating
Stricture	~3%	UAS injury, impacted stone; ultrasound surveillance (most patients never imaged → rate underestimated)
Perforation	Few %	Stop and stent
Avulsion	<1%	Proximal ureter → immediate nephrostomy, delayed reconstruction

4 · PERCUTANEOUS NEPHROLITHOTOMY

- Landmarks [MCQ]: Fernström and Johansson 1976 first percutaneous stone removal; Wickham 1984 first tubeless; Valdivia-Uría 1987 first supine; Jackman 1998 first mini-perc (13 Fr).
- Higher SFR and fewer procedures than SWL/URS for 1–2 cm; treatment of choice for **staghorn** and most stones >2 cm.

PCNL first-line	Applications
Burden	Stones >2 cm, staghorn, matrix stones
Lower pole	>1 cm; lower pole diverticulum or infundibular stenosis
Poor retrograde route	Urinary diversion, inaccessible lower tract, prior ureteral stricture/injury/reconstruction
Coexisting pathology	UPJO; infected stones in non-ambulatory patients

Preoperative and perioperative

- Detect uncorrected coagulopathy and untreated UTI. Drain first (nephrostomy/stent) after repeated complicated UTI, sepsis or pain admissions.
- Stop anticoagulants with the prescriber (bridge if high thrombotic risk); **low-dose aspirin is safe**.
- Examine habitus, spine and hip/knee flexibility for positioning. Bloods: renal function, blood count, urine culture.
- **NCCT is the gold standard**: 3 planes + 3D for branching, infundibular length/angle, SSD, HU, and ribs, pleura, spleen, liver and colon. Intraoperative pyelogram replaces prior urogram.
- Poorly functioning kidney: renogram — relative function cut-off debated at 10–20%; nephrectomy may suit a symptomatic/infected stone.
- Single-dose culture-directed prophylaxis (extended course emerging). **Tranexamic acid** (Cochrane): less transfusion and complications, no thromboembolic rise — individualise.

Positioning

Feature	Prone	Supine
Calyces	Posterior; upper pole nearest skin	Anterior easier
Airway	Better pulmonary capacity	Airway accessible

Feature	Prone	Supine
Fragments	Drain against gravity	Gravity-assisted
Retrograde access	Needs split-leg	Single position (Galdakao-modified)

- Prone: pad pressure points, arms ≤90° (brachial plexus); horizontal bolsters drop the kidney more than vertical. Risks: neuropraxia, ocular injury.

Access — the most important step

- Urologist-obtained access → shorter stay, fewer transfusions and secondary procedures. Ultrasound before draping maps ribs, diaphragm and adjacent organs.
- **Ideal**: posterior calyx with short, wide infundibulum. Upper pole posterior gives the best reach to the rest of the system, but **supracostal (above 12th rib)** raises chest complications. Lower pole: no thoracic risk but more colonic injury (left); steep infundibulopelvic angle hampers reaching the UPJ.
- Fluoroscopy under ALARA (as low as reasonably achievable): pulsed, last-image hold, saved roadmap.

Step	Eye-of-the-needle (bulls-eye)	Triangulation
Align	C-arm rotated 15–30° toward surgeon; needle aligned end-on	Mediolateral adjusted in oblique view, cephalocaudal in anteroposterior (AP)
Depth	Rotate back past AP to judge depth	Short advances alternating views

- On rotating toward the surgeon, laterally protruding posterior calyces shorten toward the spine; medially splayed (usually upper pole) calyces lengthen. **Air** fills non-dependent calyces (anterior if supine) but blocks ultrasound.
- Confirm by urine return; avoid contrast through an uncertain needle (extravasation obscures).
- **Ultrasound access**: no radiation; shows bowel, spleen, liver, pleura; 2–5 MHz curved probe; stones by shadow and twinkle artefact.
- **Endoscopic combined intrarenal surgery (ECIRS)**: retrograde ureteroscope via UAS watches the puncture, provides a target, allows through-and-through wire and visual dilatation. Fewer tracts, less bleeding, higher clearance; needs a skilled assistant.

Dilatation and tract size

- Warn the anaesthetist — dilatation is stimulating and complications start here. Safety wire beyond obstruction unless through-and-through.
- Alken metal (dense scar), Amplatz serial, or **balloon** (one pass, 20–30 atm).
- Multiple tracts raise bleeding and reduce function (especially in chronic kidney disease).

Name	Standard	Mini	Ultra-mini	Micro
Tract (Fr)	24–34	15–22	11–14	<10

- Mini vs standard: equal SFR, less bleeding/transfusion, shorter stay — but longer operation and poorer vision; advantages vanish against a 24 Fr tract.
- The nephroscope **working channel**, not the sheath, limits instruments. Warm irrigant to prevent hypothermia.

Completion and exit

- **Flexible nephroscopy/URS** at the end clears hidden calyces and ureter and inspects for injury — randomised trial: higher SFR, fewer tracts, less bleeding.

ALGORITHM

Confirm system and ureter clear → withdraw sheath under vision, safety wire last out → compress tract → persistent bleeding: re-inflate balloon or **Kaye tamponade catheter** → alert anaesthetist → **selective angioembolisation**

- Drainage: nephrostomy (small ≤10 Fr = less pain), double-J stent, or tubeless; Fibrin glue does not reduce bleeding.

Complications

- Cochrane (42 trials): **PCNL equals RIRS for major complications** and needs fewer secondary procedures.
- Access-specific: acute or delayed haemorrhage (†delayed bleeding usually from arteriovenous fistula or pseudoaneurysm → angioembolisation), infundibular stenosis, collecting system injury, bowel, splenic and hepatic injury, hydrothorax/pneumothorax (†chest drain if significant), systemic inflammatory response syndrome (SIRS)/sepsis, irrigant-related metabolic effects, venous gas embolism, deep vein thrombosis, nerve injury, function loss, death.

5 · SPECIAL ANATOMY

Anatomy	Key adaptation
Horseshoe (stones ~1 in 5)	SWL: long SSD. PCNL: access medial and caudal, upper pole posterior calyx usually subcostal; more colonic injury and access failure; flexible nephroscope for lower medial calyces. Frequent UPJO
Calyceal diverticulum/infundibular stenosis	Treat only if pain, UTI or haematuria. URS for upper/interpolar: find ostium → laser-incise or balloon dilate. Lower pole: PCNL (mini favourable). Fulguration controversial
Pelvic kidney	SWL/URS <2 cm; PCNL with ultrasound ± CT or laparoscopic guidance
Urinary diversion	Antegrade wire access most reliable; check if bladder neck closed
Transplant	Always remove stones — obstruction risks graft; denervated graft presents with fever, oliguria, haematuria, not colic. Ultrasound PCNL; dilate perigraft scar firmly
Spinal deformity	SWL poor; URS good; PCNL equal clearance but more infection; meticulous padding

- **Laparoscopic/robotic** extraction: only when SWL, URS and PCNL fail or are expected to; with UPJO, usually clear stones first then reassess the junction.

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Lower Pole I	randomised controlled trial (RCT), PCNL vs SWL, lower pole >1 cm	SFR 91% vs 21%	SWL not first-line for lower pole >1 cm
Contralateral stone RCT (2022)	Remove vs observe small asymptomatic stones during surgery	Relapse 16% vs 63%	Treat contralateral stones
Cochrane PCNL vs RIRS (2023)	42 trials	No difference in major complications; fewer secondary procedures	PCNL no longer "more morbid"

7 · MCQ ONE-LINERS

1. Asymptomatic renal stones: ~50% have an event within 5 years.
2. Stone size predicts intervention, not symptoms.
3. Treat residual fragments >4 mm after URS.
4. SWL best: density <900 HU, SSD <10 cm.
5. Patient who must stay on anticoagulant: URS.
6. Lower pole >1 cm: SWL not first-line.
7. EAU 10–20 mm: SWL and endourology in equipoise.
8. Triple D score predicts SWL, not PCNL success.
9. Modern lithotripters: electromagnetic source.
10. Piezoelectric: smallest focal volume, lowest energy.
11. Most powerful initial comminution mechanism: dynamic squeezing.
12. No routine stent with SWL.

3. First ureteral access sheath: Takayasu, 1974.
4. UAS lowers intrarenal pressure, not proven to cut infection.
5. Stent omission: normal anatomy, function, no injury, no second stage.
6. Ureteral avulsion: immediate nephrostomy, delayed reconstruction.
7. First percutaneous stone removal: Fernström and Johansson, 1976.
18. PCNL safe on low-dose aspirin.
19. Uncontrolled tract bleeding: Kaye catheter, then angioembolisation.
20. Horseshoe kidney: upper pole access usually subcostal.
21. Transplant stones present without colic (denervated graft).

99 Prevention of Urinary Stone Disease

Reading time ≈ 21 min (slow pace)

1 · STONE DISEASE AS A CHRONIC CONDITION

- Urinary stone disease (USD) is a **chronic metabolic disorder punctuated by colic**: colic is the symptom, recurrence is the disease, prevention is the long-term task. Pathogenesis → *Ch 95*.
- Recurrence up to **50%** within 5–10 years. **ROKS** (Recurrence of Kidney Stone) nomogram predicts a second stone; ROKS 2 predicts multiple recurrences. Risks: young age, male sex, high body mass index (BMI), family history, prior asymptomatic stone, brushite/struvite/uric acid stone; low urine volume and low citrate.
- Linked to the metabolic syndrome: gout and diabetes lower urine pH (uric acid stones); diabetes raises urinary calcium.
- Prevalence rising (~1 in 11 in the United States); sex ratio now near equal from a rise in young women.

2 · SCREENING EVERY FIRST-TIME FORMER

American Urological Association (AUA) and European Association of Urology (EAU) recommend a cheap triage to sort **high risk (→ 24-hour urine)** from **low risk (→ general advice)**. A 24-hour urine is not part of screening.

Component	Assess
History	Gastrointestinal (GI) disease, sarcoidosis, gout, recurrent urinary tract infection (UTI), bone disease, family history, diet; **lithogenic drugs** — calcium, vitamins C and D, acetazolamide, steroids
Blood	Sodium, potassium, creatinine, calcium (hyperparathyroidism), uric acid
Urine	pH, crystals; culture if infection suspected
Stone analysis	Uric acid → metabolic; cystine → genetic; struvite → infection
Imaging	Baseline burden (predicts events)

High risk: GI disease, obesity, metabolic syndrome, diabetes, primary hyperparathyroidism, renal tubular acidosis, sarcoidosis, gout, family history, **children**, recurrent formers.

3 · GENERAL MEASURES (ALL FORMERS)

- **Fluid** — the cornerstone for every stone type (lowers supersaturation). Target urine **≥2.5 L/day** (≈3 L intake). Adherence is poor (only ~half reach target); smart bottles and apps may help.
- **Limit nondairy animal protein**: raises urinary calcium (bone resorption, less tubular reabsorption) and uric acid, and lowers citrate (acid load). Worst when combined with low calcium intake.
- **Low sodium**: sodium raises calcium excretion and lowers citrate. Dietary Approaches to Stop Hypertension (DASH) diet lowers risk.
- **Normal calcium** intake — restriction *increases* stones (more free gut oxalate). More fruit and vegetables (alkali, citrate).

> **KEY POINT** Why normal calcium? Dietary calcium binds oxalate in the gut; restricting it frees oxalate for absorption.

4 · METABOLIC EVALUATION (HIGH-RISK/RECURRENT)

- One or two **24-hour urines** on a random diet: volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine. Two collections preferred by some (paired samples often differ).

- Fast and calcium-load testing is **discouraged** — does not change management.
- Targeted therapy matched to the urine reduces events, yet testing is badly underused.

5 · HYPERCALCIURIA — THIAZIDES

- **Preferred drug** (AUA): hypercalciuria with recurrent calcium stones, **and** recurrent calcium stones with normal evaluation. First used by Yendt (1966) MCQ.
- Mechanism: direct distal calcium reabsorption; long-term volume contraction adds proximal sodium and calcium reabsorption.
- **Potassium loss → hypokalaemia → intracellular acidosis → hypocitraturia**; so add **potassium citrate** and check potassium 1–2 weeks after starting or changing dose.
- Amiloride + thiazide: more calciuric effect, no citrate rise — **never add potassium**. Avoid triamterene (triamterene stones).
- Bonus: higher bone density, fewer hip fractures.
- Side effects: cramps, thirst, gout and diabetes (raise uric acid and glucose), sexual dysfunction; ~1 in 6 stop (ALLHAT).

Target	Drug and dose	Key side effects
Calcium	Hydrochlorothiazide 25 mg twice daily; chlorthalidone 25–50 mg; indapamide 2.5 mg	Hypokalaemia, hyperuricosuria, hypocitraturia
Citrate	Potassium citrate 30–60 mEq/day	GI upset, hyperkalaemia
Uric acid	Allopurinol 300 mg	Rash, myalgia, hypersensitivity
Oxalate	Magnesium; pyridoxine 100 mg	Diarrhoea; ataxia, numbness
Cystine	Tiopronin; D-penicillamine 250 mg titrated	See thiol table
Urease	Acetohydroxamic acid 250 mg three times daily	Heaviest burden (below)

6 · HYPEROXALURIA

- No threshold — risk rises continuously with urinary oxalate. Gut *Oxalobacter formigenes* loss (antibiotics) implicated → *Ch 95*.
- **Diet**: avoid large portions of spinach, beets, chocolate, nuts, tea; strict low-oxalate diet mainly for enteric hyperoxaluria.

	Enteric	Primary (PH)
Cause	Malabsorption, diarrhoea, Roux-en-Y gastric bypass (RYGB)	Inborn hepatic overproduction
Urine oxalate	>40, often 60–70+ mg/day; low calcium	Often >**100 mg/day**; high plasma oxalate
Onset	Adult	Childhood; nephrocalcinosis, oxalosis, renal failure
Treatment	Calcium with meals, low oxalate, fluid, liquid potassium citrate, magnesium	Pyridoxine; lumasiran (PH1)

- **Enteric mechanism**: malabsorbed fat saponifies calcium, leaving oxalate free for colonic absorption; diarrhoea adds dehydration, hypokalaemia and hypocitraturia. Restrictive bariatric procedures add less risk.
- Enteric treatment (no randomised controlled trials (RCTs)): **calcium up to 1 g/day with meals** binds gut oxalate (calcium citrate also adds alkali) — with very high fluid. Use **liquid** potassium citrate (wax tablets pass unabsorbed in rapid transit). Cholestyramine disappointing.

Type	Enzyme (gene)	Note
PH1 (~70%)	Alanine glyoxylate aminotransferase (*AGXT*)	Commonest; pyridoxine- and lumasiran-responsive
PH2 (~10%)	glyoxylate reductase/hydroxypyruvate reductase (*GRHPR*)	—
PH3 (~10%)	4-Hydroxy-2-oxoglutarate aldolase (*HOGA1*)	Mildest; often hypercalciuria

- **Pyridoxine**: cofactor raising activity and peroxisomal targeting of the PH1 enzyme.
- **Lumasiran** (Food and Drug Administration [FDA]-approved, PH1): RNA interference silencing **glycolate oxidase** → less hepatic oxalate.

7 · HYPOCITRATURIA

- Oral citrate → hepatic bicarbonate → alkali load reduces proximal citrate reabsorption. Citrate complexes calcium and inhibits crystal growth.
- **Potassium citrate** 30–60 mEq/day: AUA offers it for recurrent calcium stones with low citrate **and** with no abnormality. Sodium citrate does not lower urinary calcium (sodium load).
- Side effects: GI upset (take after meals); wax shell in stool is normal. Caution in renal impairment, potassium-sparing diuretics.
- **Over-alkalinisation (pH ≥7)** converts calcium oxalate risk into **calcium phosphate** risk; recheck 24-hour urine.
- Juices: **alkali citrate, not total citrate, matters**. Orange juice (potassium citrate) raises pH and citrate but also oxalate; lemonade (citric acid, little alkali) has most evidence and controllable sugar. Over-the-counter supplements with high total citrate may be acidic (Moonstone) [MCQ].

8 · URIC ACID STONES — A PH DISEASE

- pKa ≈ **5.5**; >90% soluble at pH 6.5. Urine pH <5.5 is the commonest cause (high acid excretion, low ammonium excretion).
- Fluid ≥2.5 L/day; limit animal protein; weight loss and glucose control.
- **Potassium citrate** to pH **6.0–6.5**; avoid >7.0. Sodium alkali risks calcium stones.
- **Allopurinol not first-line**: only for hyperuricaemia with hyperuricosuria (gout) or failed alkalinisation. Reduce dose in renal impairment; hypersensitivity syndrome (fatal in up to 27%), Stevens-Johnson syndrome; risk higher with thiazides and renal insufficiency.

9 · CYSTINE STONES

ALGORITHM

Dilute (urine ≥**3 L/day** including at night; sodium ≤2300 mg/day; moderate protein — not in children) → cystine <**250 mg/L** → **alkalinise** with potassium citrate to pH **7.0–7.5** → **thiol** (tiopronin)

- Cystine pKa 8.3 is unreachable; pH >7.5 risks calcium phosphate. Sodium alkali raises cystine excretion and blunts tiopronin.
- Thiols form a soluble cystine–drug mixed disulfide.

	D-Penicillamine	Tiopronin	Captopril
Status	First thiol; largely replaced	**Thiol of choice**; fewer stop	Weak evidence
Dose	250 mg/day, titrate	300 mg twice daily, up to 1200 mg/day	—
Toxicity	Rash, fever, arthralgia, pancytopenia, nephrotic syndrome	Asthenia, GI upset, rash, joint aches	—

10 · INFECTION (STRUVITE) STONES

- **Surgical disease first**: complete clearance — percutaneous nephrolithotomy (PCNL) for staghorn → *Ch 98* — since every fragment harbours bacteria and endotoxin.
- Urine and stone cultures correlate poorly → broad-spectrum cover at surgery; then long-term infection control and drainage.
- **Acetohydroxamic acid (AHA)**: urease inhibitor; prevents new stones and growth. AUA: only for residual/recurrent struvite after surgical options exhausted. Toxicity in ~30%: haemolytic anaemia, **deep venous thrombosis**, tremor, headache. Contraindicated with creatinine **>2.5 mg/dL** and pregnancy.
- Most struvite formers also have hypercalciuria/hypocitraturia → do a 24-hour urine.

11 · MONITORING

ALGORITHM

Start therapy → 24-hour urine within **6 months** → then at least yearly urine + blood for side effects → new stone: **re-analyse** → periodic imaging

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Borghi 1996	RCT, high fluid, calcium oxalate formers	5-year recurrence 12% vs 27%	Urine >2 L/day prevents stones
Borghi 2002	RCT, hypercalciuric formers	Low protein/salt, normal calcium halved events vs low calcium	Never restrict calcium
Ettinger 1988	Double-blind RCT, chlorthalidone	~90% fewer stones	Thiazides prevent calcium stones
Barcelo 1993	RCT, potassium citrate	Fewer new stones at 3 years	Citrate prevents recurrence
ILLUMINATE-A	Phase 3 RCT, lumasiran, PH1	84% vs 0% reached near-normal urine oxalate	RNA interference for PH1

13 · MCQ ONE-LINERS

1. Most important single risk factor: low urine volume.
2. Target urine output ≥2.5 L/day.
3. Calcium restriction increases stone risk.
4. 24-hour urine is not part of first-stone screening.
5. Thiazide hypocitraturia arises via hypokalaemia.
6. Amiloride plus thiazide: never add potassium.
7. Enteric hyperoxaluria: calcium with meals.
8. Enteric hyperoxaluria: liquid potassium citrate.
9. PH1 gene: AGXT; commonest type.
0. Lumasiran targets glycolate oxidase.
11. Sodium citrate does not lower urinary calcium.
12. Urine pH ≥7 favours calcium phosphate stones.
13. Orange juice raises urinary oxalate.
14. Uric acid pKa 5.5.
15. Uric acid target pH 6.0–6.5.
16. Allopurinol: not first-line for uric acid stones.
17. Cystine target <250 mg/L; pH 7.0–7.5.
18. Thiol of choice: tiopronin.
19. AHA contraindicated if creatinine >2.5 mg/dL.
20. AHA's most severe toxicity: deep venous thrombosis.
21. Repeat 24-hour urine within 6 months of therapy.

100 Lower Urinary Tract Calculi

Reading time ≈ 13 min (slow pace)

1 · BLADDER CALCULI: CLASSIFICATION AND AETIOLOGY

- Commonest lower tract stone (~5% of urinary stones); declining in the West, persisting in developing nations, especially children.
- Types: **migrant**, **primary** (idiopathic/endemic) and **secondary** (predisposing cause).

	Primary (endemic)	Secondary
Patient	Children; peak **2–4 years**; male ≫ female	Men >60 with outlet obstruction
Cause	Cereal diet low in animal protein and phosphate → hypophosphaturia	Bladder outlet obstruction (BOO), foreign body, catheter, infection
Composition	Ammonium acid urate, calcium oxalate, uric acid	**Struvite**
Recurrence	Solitary; rarely recurs	Recurs unless cause removed

- Secondary causes: significant post-void residual (PVR) — stricture, benign prostatic hyperplasia (BPH), neurogenic bladder, diversion; **foreign bodies** — migrated intrauterine device, eroded sutures, burst Foley balloon fragments, hair during clean intermittent catheterisation (CIC). Mechanism: **heterogeneous nucleation** on a nidus.
- Drugs: triamterene, indinavir.

> **KEY POINT** Primary stone is a disease of the urine (normal bladder); secondary is a disease of the bladder (stasis, nidus, infection). Do not hunt for outlet obstruction in the endemic child.

2 · PRESENTATION

- Adults: **terminal haematuria** with lower urinary tract symptoms (LUTS) — intermittency, poor flow, urgency; occasionally acute retention.
- Children: cloudy, sandy urine; dysuria, frequency, abdominal discomfort; retention rare.

3 · MANAGEMENT OF BLADDER STONES

Choice depends on size, composition, age, build, coexistent stones, BOO and expertise. **The route, not the stone, dictates modality.**

Option	Indication / key point
Chemodissolution	Adjunct only; best for catheter encrustation
Extracorporeal shock wave lithotripsy (ESWL)	Artificial sphincter, penile prosthesis, neobladder, high-risk patient; poor in continent reservoirs
Transurethral cystolithotripsy	Most popular; holmium laser of choice (single stage); electrohydraulic lithotripsy (EHL), pneumatic. Risk: urethral stricture
Percutaneous cystolithotomy	Difficult urethra (bladder neck reconstruction/closure); Amplatz sheath, entrapment sac; spares urethra
Open cystolithotomy	Small-capacity bladder, stricture, no safe endoscopic route

- Terms: cystolithotomy = intact removal; cystolithotripsy = energy fragmentation; cystolitholapaxy = mechanical crushing (lithotrite — mucosal injury, perforation).

4 · SPECIAL SITUATIONS

BOO

- BOO in 45–79% of bladder stones. **Bladder stone secondary to BOO is an absolute indication for BPH surgery**.

- But >30% of stones are infected (urea-splitters) and some patients have no BOO → **prove BOO**, do not assume it.

Diversion and augmentation

- Metabolic: hyperchloraemic acidosis → hypercalciuria; ileal loss → enteric hyperoxaluria. Infection: urease producers (*Proteus*, *Klebsiella*, *Pseudomonas*) → struvite; **mucus** fosters biofilm. Stasis. Absorbable sutures reduce stones.

Reconstruction	Approach
Conduit with stomal stenosis	Clear stone + revise stoma
Continent cutaneous pouch	Percutaneous — protects continence mechanism
Neobladder, small/large burden	Transurethral / percutaneous or open

- Know the anatomy preoperatively (avoid bowel and pedicle injury).

Spinal cord injury (SCI)

- Risk peaks **3 months** after injury; 15–30% within 10 years; quadruples after first stone. Acute phase: immobilisation hypercalciuria (calcium stones); later: infection.
- **CIC beats indwelling catheters**; suprapubic = urethral. If indwelling unavoidable, weekly changes.
- Treat: ≤4 cm transurethral laser; larger percutaneous/open; ESWL for anaesthetic risk or autonomic dysreflexia.

Transplant and children

- Transplant: *Proteus*, outflow obstruction, retained suture; routine treatment — **remove all exposed suture**.
- Children: the **small urethra** is the constraint — percutaneous cystolithotripsy (PCCL) for large stones, even in infants; transurethral laser for smaller.

5 · URETHRAL CALCULI

- 0.3–1% of stones; males; mostly **migratory** from the upper tract, arrested by stricture, meatal stenosis or BPH. Commonest site: posterior urethra (up to 88%).

	Migratory	Primary
Frequency	Vast majority	Minority — stricture, diverticulum, hair-bearing graft, foreign body
Composition	Calcium oxalate	Struvite
Course	Acute	Insidious, years; infection

- Presentation: **acute painful retention**; posterior → perineal/rectal pain; anterior → penile-tip pain, palpable mass. Delay → urethrocutaneous/urethrorectal fistula.
- ~98% radio-opaque → plain X-ray; retrograde urethrogram (RUG) if ultrasound inconclusive or stricture suspected; image upper tract.

Site	Treatment
Posterior	**Push back** to bladder → laser; else in situ laser/pneumatic
Anterior	**No pushback**; in situ holmium laser; milk only smooth stones; urethrotomy for large
Fossa navicularis	Forceps; ventral meatotomy
Diverticular / stricture	Extraction + diverticulectomy or stricture repair, same or staged

6 · UNUSUAL SITES

- **Preputial**: severe phimosis, poor hygiene; smegma and urinary stasis; struvite → circumcision or dorsal slit, **excised skin to histopathology**.

- **Prostatic**: prostatic-urethral stones are migratory (manage as urethral); true calculi lie in acini (endogenous, small, many) or form by urine reflux (exogenous, larger). Calcium phosphate. Linked to BPH, **not cancer**; prostatitis symptoms → 3.2× likelihood. Transrectal ultrasound (TRUS) main tool; plain X-ray detects 14%. Remove with loop at transurethral resection of prostate (TURP).
- **Seminal vesicle**: haematospermia, painful ejaculation; TRUS most accurate. Standard: vesiculectomy or transurethral resection of ejaculatory ducts (TURED); transurethral seminal vesiculoscopy (6 French, via verumontanum) has fewer complications.

7 · MCQ ONE-LINERS

1. Endemic bladder stone peak: 2–4 years.
2. Endemic stone composition: ammonium acid urate.
3. Secondary bladder stone composition: struvite.
4. Commonest bladder stone presentation: terminal haematuria.
5. Foreign-body stones: heterogeneous nucleation.
6. Bladder stone from BOO: absolute indication for BPH surgery.
7. Laser: modality of choice for transurethral cystolithotripsy.
8. Percutaneous cystolithotomy spares the urethra.
9. ESWL preferred with artificial sphincter or penile prosthesis.
10. Continent pouch stone: percutaneous approach.
11. SCI stone risk peaks 3 months post-injury.
12. CIC lowers stone risk versus indwelling catheter.
13. Transplant bladder stone: remove exposed suture.
14. Urethral calculi: 0.3–1% of stones.
15. Commonest urethral stone site: posterior urethra.
16. Urethral stone presentation: acute painful retention.
17. Anterior urethral stone: never push back.
18. Primary urethral stone: struvite, from stricture or diverticulum.
19. Prostatic calculi: no link with adenocarcinoma.
20. Preputial calculi: send prepuce for histology.
21. Seminal vesicle stones: TRUS most accurate.

54 Management of Paediatric Kidney Stone Disease

Reading time ≈ 25 min (slow pace)

1 · EPIDEMIOLOGY: THE ADULT DISEASE STARTED EARLY

- Early-onset disease = onset < **24 years**. Incidence rising fastest in **adolescent girls**. About 50% recur within 3–5 years (≥ adult rate).
- Determinants: polygenic susceptibility × exposome (low fluid, high sodium, drugs) × comorbidity.
- Oral antibiotics raise stone risk for up to 5 years, most in children (loss of oxalate-degrading gut bacteria).

Feature	Children	Adults
Sex ratio	Female predominance	Male (historically 3 : 1)
Calcium oxalate (CaOx)	≈ **80%** — mostly idiopathic, on Randall's plaques, not genetic or infective	75–80%
Calcium phosphate (CaP) / uric acid	CaP slightly commoner, uric acid rarer — child's urine is more alkaline	≈ 5% each
Lifetime stakes	Longer runway for recurrence, chronic kidney disease (CKD), hypertension, low bone mineral density (BMD); more cumulative radiation and anaesthesia	Shorter

- **High-risk groups**: augmentation cystoplasty (≈ 1%/year — acidosis, mucus nidus, colonisation); malabsorption; stasis; tube feeds (high sodium/oxalate, low calcium); non-ambulatory (bone resorption → hypercalciuria, infection stones).

> **KEY POINT** Stone formers have more CKD, hypertension and fracture; childhood is the bone-accrual window, so seek low BMD early.

2 · DIAGNOSIS AND IMAGING

- Older children present like adults (flank pain, vomiting, lower urinary tract symptoms); **young or non-verbal children cannot localise pain** → imaging and laboratory tests carry the diagnosis. Always ask about fever.

ALGORITHM
Suspected stone → **ultrasound (US)** grey-scale + colour Doppler → positive: treat as indicated → negative + high suspicion or hydronephrosis → **low-dose non-contrast computed tomography (NCCT)** → negative + low suspicion: stop.

- US stone = hyperechoic focus + **twinkle artefact** on Doppler (maximise pulse repetition frequency). Shadowing raises specificity but is often absent for stones < 4 mm.
- US sensitivity > 70%, specificity > 95%; overestimates size by 1–2 mm. Missed stones are usually small and non-obstructing; ureteric stones are poorly seen.
- NCCT ≈ 100% accurate, but children carry more radiation risk (longer life, radiosensitive tissue) → reserve computed tomography (CT) for a suspected obstructing ureteric stone unseen on US, where a miss costs renal function. Use 40 mA (half dose) for ≤ 50 kg.

3 · METABOLIC EVALUATION — EVERY CHILD

An identifiable metabolic abnormality raises recurrence risk up to **fivefold**, and early onset implies more monogenic disease, a more severe phenotype and longer exposure — hence **comprehensive evaluation after the first stone**.

- **History**: family history; diet (fluid, salt, vitamins C/D, ketogenic); drugs (steroids, furosemide, acetazolamide, topiramate); prematurity, anomalies, urinary tract infection (UTI), immobility.
- **Examination clues**: dysmorphism → Williams; rickets → Dent, hereditary hypophosphataemic rickets with hypercalciuria (HHRH); tetany → familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC)).
- **Stone analysis** first (infrared spectroscopy or X-ray diffraction): cystine, struvite, 2,8-dihydroxyadenine (2,8-DHA) and xanthine are diagnostic.
- **Serum**: creatinine, calcium, phosphorus, bicarbonate, magnesium, uric acid. Parathyroid hormone only with hypercalcaemia + hypophosphataemia. Hypercalcaemia + hypercalciuria → 25-hydroxy : 24,25-dihydroxyvitamin D ratio for **CYP24A1** loss. Males with recurrent calcium stones + proteinuria → urine β2-microglobulin (Dent).
- **24-hour urine**: calcium, oxalate, uric acid, sodium, citrate, creatinine, volume, pH, cystine; index to weight, body surface area (BSA) or creatinine. Not toilet-trained → spot solute : creatinine ratios.

Metabolite	Normal (child)
Calcium	< **4 mg/kg/day**; spot Ca : creatinine < 0.8 (0–6 mo), < 0.6 (7–12 mo), < **0.21** (> 2 y)
Oxalate	< 40–50 mg/1.73 m^2/day; spot ratio falls with age
Citrate (lower limit)	Hypocitraturia < 128 mg/g creatinine (male), < 300 (female)
Cystine	< 50–60 mg/1.73 m^2/day
Uric acid	< 815 mg/1.73 m^2/day; spot < 0.56 mg/dL glomerular filtration rate (GFR)

> **KEY POINT** Never read a child's urine against adult absolute limits: infants physiologically excrete more calcium, so the school-age 0.21 ratio overcalls hypercalciuria. Citrate is an inhibitor — its abnormality is too *little*.

4 · URINARY ABNORMALITIES

Hypercalciuria — commonest (30–50%)

Idiopathic hypercalciuria is diagnosed only after secondary causes are cleared. Serum calcium sorts them:

Serum calcium	Causes
High	Hyperparathyroidism, CYP24A1, hypervitaminosis D, immobilisation, Williams
Low	Hypoparathyroidism; autosomal dominant hypocalcaemic hypercalciuria (gain-of-function calcium-sensing receptor)
Normal	Prematurity, loop diuretics, topiramate, ketogenic diet, Dent, Bartter, FHHNC, distal renal tubular acidosis (dRTA), HHRH → else **idiopathic**

Hyperoxaluria

- **Primary hyperoxaluria (PH)**: autosomal recessive hepatic overproduction → nephrocalcinosis, stones, CKD; phenotype ranges from neonatal renal failure to adult stones.
- Raised oxalate → urinary glycolate (PH1), L-glycerate (PH2), 4-hydroxy-2-oxoglutarate (PH3) → genetics; all normal → secondary.
- **Secondary**: fat malabsorption (inflammatory bowel disease, cystic fibrosis, short bowel) — fatty acids bind luminal calcium, freeing oxalate; low-calcium diet; vitamin C.

Type	Gene / enzyme	Note
PH1	*AGXT* / alanine–glyoxylate aminotransferase (AGT; pyridoxine cofactor)	Most severe; pyridoxine trial
PH2	*GRHPR*	Milder
PH3	*HOGA1* (mitochondrial)	—

Others

- **Hypocitraturia**: proximal tubular acidosis (metabolic acidosis, hypokalaemia) increases citrate reabsorption — ketogenic diet, topiramate, acetazolamide, dRTA, diarrhoea. Incomplete dRTA needs acid-load testing.
- **Cystinuria**: autosomal recessive, *SLC3A1*/*SLC7A9*; cystine, ornithine, lysine, arginine lost, only cystine crystallises. May present in infancy with staghorn stones.
- **Hyperuricosuria**: children excrete more urate, yet uric acid stones < 5% (alkaline urine); urate seeds CaOx by epitaxy. Causes: hypoxanthine–guanine phosphoribosyltransferase (HPRT) deficiency (Lesch–Nyhan), phosphoribosyl pyrophosphate synthetase (PRPS) superactivity, renal hypouricaemia (*SLC22A12*), purine excess.

Monogenic disease	Inheritance	Signature
Dent (*CLCN5*)	X-linked	Low-molecular-weight proteinuria, nephrocalcinosis, CKD
Bartter	Recessive	Salt wasting, hypokalaemic **alkalosis**, hypercalciuria
FHHNC (*CLDN16/19*)	Recessive	Hypomagnesaemia, tetany, nephrocalcinosis
dRTA	Dominant/recessive	Hypokalaemic **acidosis**, hypocitraturia, nephrocalcinosis

5 · MANAGEMENT PRINCIPLES

- **Medical expulsive therapy (MET)**: α-blockers relax distal ureter (α1A/α1D) → more passage in children (off-label tamsulosin, ≤ 6 weeks).
- Up to 60% need surgery. Beyond stone size/site and anatomy, children's choices weigh **number of anaesthetics** (neurotoxicity concern before school age) and **cumulative radiation** (pulsed low-dose fluoroscopy, collimation, US guidance).
- Residual fragment threshold: **4 mm on US** (2 mm CT threshold + ≈ 2 mm US overestimate).
- Urine culture before every procedure; prophylaxis for all ureteroscopy (URS) and percutaneous nephrolithotomy (PCNL), shockwave lithotripsy (SWL) only if infection risk; stop within 24 h.

Scenario (American Urological Association / Endourological Society 2016)	Recommendation
Uncomplicated ureteric ≤ **10 mm**	Observation ± α-blocker
Ureteric, failed/unlikely to pass	URS or SWL (strong)
Before URS	No routine pre-stenting
Renal ≤ **20 mm**	SWL or URS
Renal > 20 mm	PCNL or SWL (with stent/nephrostomy); authors: PCNL
Before PCNL	Low-dose CT (strong)
Asymptomatic non-obstructing renal	Surveillance with US
Open/laparoscopic/robotic	Only with anatomic abnormality

6 · URETEROSCOPY

- Clearance ≈ 90%; worse in children < 6 years or < 20 kg and for lower-pole stones.
- Semirigid 4.5–6 French (Fr) (ureter); flexible 7.5–9 Fr (proximal ureter, kidney). Lower pole: 150–200 µm fibre (preserves deflection).

ALGORITHM

General anaesthesia + sterile culture → padded lithotomy (infants: gel rolls) → cystoscopy → retrograde pyelogram + safety wire → calibrate orifice with 8/10 Fr coaxial dilator → **if it will not pass: 4.6–4.8 Fr stent ≥ 14 days, return later** → URS → stent if long case, many passes, trauma/oedema.

- **No balloon or active dilatation** of the paediatric orifice (stricture, perforation): passive dilatation by stent is preferred to force.
- Warm isotonic irrigant (avoids hyponatraemia, hypothermia).
- Complications: ureteric injury → abort and stent. Stricture → laser incision (not balloon) or reconstruction by level.

7 · SHOCKWAVE LITHOTRIPSY

- Child advantage: **short skin-to-stone distance**; fewer anaesthetics than URS, equal clearance for small stones.
- Favourable: < 10 mm, non-lower-pole, < 1000 Hounsfield units (HU), short infundibulum. Unfavourable: cystine, brushite, infundibulopelvic angle > 45°.
- Technique: usually general anaesthesia; < 20 kV; ≤ 2000–3000 shocks (< 2000 in infants); **slower rate** (80 better than 120/min); foam lung shielding; ungated shocks safe.
- Absolute contraindications: bleeding diathesis, untreated UTI, pregnancy. Complications: haematoma, steinstrasse.

8 · PERCUTANEOUS NEPHROLITHOTOMY

- First-line > 20 mm; clearance ≈ 90%. A 24–30 Fr tract in a child ≈ 72 Fr in an adult → **miniaturisation** (mini < 20–24 Fr; micro < 10 Fr) reduces bleeding.
- Planning: CT (calyx, adjacent organs, spinal deformity); treat UTI beforehand; only where paediatric interventional radiology can embolise.

ALGORITHM
Hypothermia prevention → ureteric catheter → prone (latex precautions in myelomeningocele) → US/fluoroscopic puncture of posterior calyx → serial dilators → 18 Fr sheath (24 Fr staghorn; trim in small children) → nephroscope + ultrasonic/laser → flexible check → nephrostomy (7 Fr infant to 16 Fr adolescent).

- Infant kidneys are mobile and push away during dilatation.
- Transfusion < 10%. Bleeding → abort, tamponade with catheter; late bleeding (pseudoaneurysm) → angioembolisation. Pelvic perforation → stop, antegrade stent.

9 · SECONDARY PREVENTION

- **Fluid**: ≥ maintenance (children); > 2–2.5 L/day (adolescents); 1.5–2 L/m² in cystinuria or PH.
- **Sodium** < 2–3 mEq/kg/day — sodium drives calciuria and cystinuria. **Calcium not restricted** (binds gut oxalate; protects bone). Animal protein to daily allowance; potassium-rich produce; limit oxalate and vitamin C.
- Urine pH target: > 6.5 uric acid, > 7 cystine; avoid > 6.5 in calcium stone formers (CaP).

Drug	Indication	Child dose / caution
Hydrochlorothiazide	Hypercalciuria despite salt restriction	1–2 mg/kg/day; hypokalaemia → add potassium citrate
Potassium citrate	Hypocitraturia, uric acid, cystine	2–4 mEq/kg/day; CaP if pH > 6.5
Tiopronin	Cystinuria failing fluid + alkali	Thiol; fewer adverse effects than penicillamine
Allopurinol	HPRT, PRPS, adenine phosphoribosyltransferase deficiency; hyperuricosuric CaOx	4–10 mg/kg/day; xanthine stones
Pyridoxine	PH1 (trial in all)	10–30% respond
Lumasiran	PH1	RNA interference against glycolate oxidase

For uric acid stones, fluid + alkali, not allopurinol, is the mainstay. † PH1 with end-stage renal disease: combined liver–kidney transplantation corrects the hepatic enzyme defect; kidney-alone grafts are lost to recurrent oxalosis.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ILLUMINATE-A	Randomised, lumasiran vs placebo, PH1	Urinary oxalate ↓ 65%	RNA interference therapy for PH1

11 · MCQ ONE-LINERS

1. Early-onset stone disease: onset before 24 years; female predominant.
2. Commonest paediatric stone: calcium oxalate (≈ 80%).
3. Alkaline paediatric urine: more CaP, fewer uric acid stones.
4. First imaging: US; twinkle artefact confirms stone.
5. Commonest metabolic abnormality: idiopathic hypercalciuria (30–50%).
6. Hexagonal crystals = cystine.
7. PH1 gene *AGXT*; urinary glycolate raised.
8. Dent disease: X-linked *CLCN5*, low-molecular-weight proteinuria.
9. Bartter = alkalosis; dRTA = acidosis; both hypercalciuric.
10. CYP24A1 loss mimics vitamin D intoxication.
11. α-Blocker MET for ureteric stones ≤ 10 mm.
12. No routine pre-stenting before paediatric URS.
13. Tight ureter: stent ≥ 14 days, no balloon dilatation.
14. Renal ≤ 20 mm: SWL or URS; > 20 mm: PCNL.
15. Low-dose CT before PCNL is a strong recommendation.
16. Significant residual fragment on US: 4 mm.
17. Slower SWL rate (80/min) clears better than 120/min.
18. Calcium is not restricted — it binds gut oxalate.
19. Thiols and lumasiran: only drugs approved for paediatric prevention.
20. Pyridoxine: trial in every PH1 patient.

125 Genital and Lower Urinary Tract Trauma

Reading time ≈ 32 min (slow pace)

1 · PRINCIPLES

- Lower tract injuries are rarely life-threatening; long-term disability follows mismanagement, not the injury.
- Key decision — **immediate reconstruction vs deliberate delay**: testicular rupture → immediate repair; pelvic fracture urethral injury (PFUI) → suprapubic tube now, urethroplasty later.
- Imaging localises the injury: ultrasonography for scrotum and equivocal penile fracture; computed tomography (CT) cystography for bladder; retrograde urethrography (RUG) for urethra.

2 · PENILE FRACTURE

Anatomy and mechanism

- Definition: **tear of the tunica albuginea** with corpus cavernosum rupture. Tunica is bilaminar (inner circular, outer longitudinal); outer layer gives strength and is **thinnest ventrolaterally**.
- Bending of the **erect** penis → abrupt rise in intracavernosal pressure → transverse tear of proximal–mid shaft, usually ventral or lateral. Bilateral in ~10% — bilateral tears carry more urethral injury.
- Causes: intercourse (rear-entry commonest); in the Middle East self-inflicted "taqaandan" (forcible bending for detumescence). Collagenase clostridium histolyticum (CCH) for Peyronie disease — most ruptures occur after the 14-day observation window.

Diagnosis

- **Clinical diagnosis** (American Urological Association [AUA] Urotrauma Guideline): "snap/pop" → pain → rapid detumescence → swelling and discoloration. Penis deviates **away** from the tear; defect may be palpable.
- **Buck's fascia** decides haematoma pattern: intact → confined to shaft ("eggplant deformity"); breached → tracks to scrotum, perineum and suprapubic region †(butterfly pattern).
- Urethral injury ~20% in the West vs ~3% in Asia/Middle East (intercourse vs self-inflicted). Blood at meatus, haematuria or retention suggest it, but their absence does not exclude it.
- Urethral evaluation: **intraoperative flexible cystoscopy** before catheterisation is now routine (AUA; EAU [European Association of Urology] prefers it to RUG).
- Imaging only if equivocal: **ultrasonography** first (rules out, localises tear for incision); magnetic resonance imaging (MRI) if ultrasound equivocal — a negative MRI reliably excludes fracture. Cavernosography discouraged.
- Mimics: "false fracture" and dorsal penile artery/vein rupture — may reproduce the history; explore or MRI.

Management

ALGORITHM
Prompt exploration → ventral penoscrotal (or small lateral) incision → evacuate haematoma → identify tear (artificial erection with saline/dye) → irrigate → close tunica with interrupted **2-0/3-0 absorbable** → repair urethra if injured

- Distal circumcising incision risks preputial ischaemia in the uncircumcised — add circumcision at the end if used. Avoid deep corporal ligation and excessive debridement of erectile tissue.
- Urethra: partial → oversew over catheter; complete → debride, mobilise, tension-free anastomosis. Catheter 2–3 weeks, antibiotics, 1 month abstinence.
- Surgery beats conservative care: less curvature, erectile dysfunction (ED) and morbidity; complications ~20% (nodules, curvature, ED). **Delay up to 7 days does not worsen outcome** — still explore.
- **CCH-related fracture**: trend to conservative care (outcomes equivalent) unless urethral injury.

3 · PENETRATING, BITE, AMPUTATION, ZIPPER AND STRANGULATION INJURIES

- **Gunshot**: explore immediately (do not delay for imaging) → irrigate → remove foreign matter → antibiotics → close tunica; artificial erection confirms closure. Almost never isolated. Urethral injury in **15–50%** → RUG or intraoperative dye injection. Viable tissue → primary urethral repair; high-velocity/shotgun tissue loss → suprapubic diversion and staged repair.
- **Dog bites**: irrigate → debride → **primary closure** → broad-spectrum antibiotic (amoxicillin–clavulanate); tetanus and rabies cover. **Human bites**: present late and infected — usually **leave open**; amoxicillin–clavulanate or moxifloxacin.

Penile amputation

- Usually self-mutilation → **psychiatric consultation always**. Preferred: **microsurgical replantation**; transfer if available, else macroscopic repair of urethra and corpora.
- **"Double bag"**: saline-soaked gauze → sterile bag → outer bag with ice slush; inner bag prevents direct ice injury. Long ischaemia time alone is no reason to abandon replantation.
- Erections in >50% with either technique; microvascular repair has less stricture, skin loss and sensory loss. Skin behaves as a free graft unless superficial vessels are repaired → necrosis → later split-thickness graft.
- Part unavailable → formalise stump as in partial penectomy. Total loss → radial forearm free flap phalloplasty.

ALGORITHM
Suprapubic cystostomy → debride → tunica with 2-0 absorbable → urethra over catheter 4-0 → minimal neurovascular dissection → dorsal artery **11-0**, dorsal vein **9-0**, dorsal nerve **10-0** nylon → skin cover

- **Zipper** (children): penile block → mineral oil → one attempt to unzip; else cut cloth between teeth, or cut the slider's **median bar** with bone cutters so it falls apart; circumcision under general anaesthesia if needed.
- **Strangulation**: in children, think of hair tourniquet and **child abuse**. Decompress urgently: lubricate and remove → **string technique** (tape passed under ring, wound distally, unwound from proximal end) → glanular puncture to release blood → cutting tools (saw, industrial drill) with protection and saline cooling, in theatre. Suprapubic catheter if retention.

4 · TESTICULAR INJURY

- Protected by scrotal mobility, cremasteric reflex and tough tunica. Blunt (assault, sport) or penetrating; ~80% of penetrating scrotal wounds involve neighbouring structures (thigh, penis, urethra, **femoral vessels**). Blast triad: scrotal skin loss, bilateral testicular fragments, limb/abdominal destruction.
- Consider rupture in every blunt scrotal injury. Haematoma size does not predict rupture.
- **Ultrasonography**: heterogeneous parenchyma, tunical disruption; sensitivity only ~64–75% (60% in gunshot) — **normal or equivocal scan must not delay exploration** when examination suggests injury.
- Non-palpable testis after trauma (motorcycle) → **dislocation**, commonest to superficial inguinal pouch (50%) → reduction. Trauma does not exclude torsion.

†American Association for the Surgery of Trauma (AAST) testis grade	Injury
I	Contusion/haematoma
II	Subclinical laceration of tunica albuginea
III	Tunical laceration with <50% parenchymal loss

†American Association for the Surgery of Trauma (AAST) testis grade	Injury
IV	Major laceration with ≥50% parenchymal loss
V	Total destruction or avulsion

ALGORITHM
Explore (transverse or vertical scrotal incision) → remove necrotic/extruded tubules → close tunica with small absorbable sutures → large defect: tunica vaginalis flap/graft (no synthetic graft) → drain

- Close even small defects — rising intratesticular pressure keeps extruding tubules.
- Explore **significant haematocele** (up to 80% hide rupture) and **intratesticular haematoma** (else atrophy; delayed orchidectomy 15%).
- Penetrating → always explore; ligate injured vas with non-absorbable suture, reconstruct later; consider contralateral exploration.
- Salvage **>90%** if repaired within **72 h**; non-operative care → infection, atrophy, pain, late orchidectomy. Solitary testis: protective cup, sport not restricted.

5 · GENITAL SKIN LOSS

- Causes: Fournier gangrene debridement (commonest), machinery/suction avulsion (loose areolar plane strips cleanly), burns (penile burns usually full thickness), blast.
- Fournier: crepitus, skin ischaemia, lost scrotal rugae; subcutaneous air on ultrasound/CT. Treat with antibiotics, resuscitation and **repeated debridement**; wet-to-dry or negative pressure wound therapy (NPWT), still inspecting daily; early suprapubic diversion in extensive injury. Hyperbaric oxygen unproven.
- Deep electrical burns: conservative — outcome dictated by concurrent injuries.

	Penis	Scrotum
Graft	**Thick, non-meshed** split-thickness skin graft (STSG) 0.012–0.015 inch (meshed contracts)	**Meshed** STSG (exudate escapes, better take) after local flaps
Key step	Excise residual subcoronal skin (else lymphoedema); never reuse avulsed skin	Suture testes together to avoid bifid scrotum; ≤50% loss closes primarily
Note	Shaft graft insensate but sexual function preserved	Thigh pouches not recommended

- Graft 5–7 days after last debridement (or ≥72 h culture-directed antibiotics, granulation, normal white count); bolster or vacuum dressing.

6 · BLADDER INJURY

Aetiology and presentation

- Deep pelvic position → injury implies great force or a full bladder. Blunt: pelvic fracture (bony laceration or shearing at fascial attachments); blow to full bladder → **dome** rupture. Iatrogenic: obstetric/gynaecological surgery commonest.
- **Pelvic fracture present in 83–95%** of bladder injuries, but bladder injury in only **5–10%** of pelvic fractures. Mortality is from associated injuries.
- **Gross haematuria** in nearly all; suprapubic pain, inability to void, free fluid, ileus.
- Extraperitoneal ↔ pelvic fracture; intraperitoneal ↔ full-bladder blow or penetrating injury.

†AAST bladder grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm

†AAST bladder grade	Injury
IV	Intraperitoneal laceration ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

Imaging

- **Absolute indication** for cystography: **gross haematuria + pelvic fracture** (~29% have rupture). Relative: gross haematuria alone, microhaematuria + pelvic fracture — decide on clinical indicators. Penetrating pelvic/buttock injury with any haematuria → cystography.
- Retrograde (stress) cystography ~100% accurate: fill to discomfort or **350 mL**; scout → full → **drainage film** (else posterior leak missed). Underfilling gives false negatives.
- **CT cystography** equally accurate: retrograde fill with contrast diluted to 2–4% (6:1 with saline) to 350 mL; no drainage film needed. Clamping the catheter during intravenous-contrast CT is inadequate — retrograde filling is mandatory.
- Extraperitoneal: **flame-shaped** pelvic contrast. Intraperitoneal: contrast outlines bowel loops.

Management

- **Uncomplicated extraperitoneal → 22-Fr urethral catheter** alone; cystogram before removal at 2–3 weeks; repair if not healed by 4 weeks. Bone spicules in the bladder wall prevent healing → repair. Poor drainage = catheter problem until proven otherwise.
- Repair extraperitoneal injury when at laparotomy anyway (open dome in midline, close intravesically, leave pelvic haematoma) or with complicating features.
- **Intraperitoneal and penetrating** → immediate repair: tears are larger than imaging suggests and will not heal; leak → chemical peritonitis, and urine reabsorption causes **acidosis and azotaemia**. Laparoscopic repair in selected isolated cases. Cystogram at 7–10 days.
- Check ureteric efflux; orifice injury → stented reimplant. **Suprapubic tube adds nothing** over urethral catheter.
- Rectal/vaginal injury: separate walls, non-overlapping suture lines, interpose viable tissue.

> **KEY POINT** Indications for immediate repair: intraperitoneal, penetrating, bladder neck, rectal/vaginal injury, bone fragments in bladder, clots/poor drainage, iatrogenic urological injury, and laparotomy for other reasons.

7 · POSTERIOR URETHRAL INJURY (PFUI)

- Anterior ring fracture or symphyseal diastasis; highest risk with **straddle (four-rami) fractures** and vertically/rotationally unstable pelvis. ~10% of male and up to 6% of female pelvic fractures.
- Puboprostatic ligaments fix the urethra; disruption occurs at the **bulbomembranous junction** — sphincter complex usually preserved, so continence often maintained.
- Triad: **blood at meatus, inability to void, palpable bladder**; "high-riding" prostate is unreliable (haematoma). Females: vaginal examination mandatory in every pelvic fracture; urethroscopy preferred to RUG.

†AAST urethra grade	Injury
I	Contusion — blood at meatus, normal urethrogram
II	Stretch — elongation without extravasation
III	Partial disruption — extravasation, contrast reaches bladder
IV	Complete disruption, <2 cm separation
V	Complete, >2 cm separation or extension into prostate/vagina

Retrograde urethrography

ALGORITHM
Blood at meatus → RUG before catheter: 16-Fr catheter in fossa navicularis, balloon 3 mL → penile traction → oblique position → inject 20 mL contrast → image during injection

Acute management

- **Suprapubic tube** is standard: percutaneous (Trendelenburg → spinal needle → peel-away sheath → 16-Fr Foley; avoid small "punch" tubes) or open (inspect bladder). Suprapubic tube is safe with pelvic hardware; tunnel it high in midline away from the plate.
- Alternative: **brief primary realignment** in stable patients (combined antegrade/retrograde flexible cystoscopy). No difference in stenosis or urethroplasty need vs suprapubic tube; prolonged attempts risk infection. Remove urethral catheter at 4–6 weeks but keep the suprapubic tube, because most still develop stenosis; remove it 7–14 days later if voiding well.
- Partial tears → gentle catheterisation is acceptable (does not complete the tear); confirm position radiographically.

Delayed reconstruction

- Defect fills with scar — a **stenosis**, not a stricture. Repair at **2–3 months** once ambulant.
- Work-up: simultaneous cystogram and RUG ("up-and-down-o-gram") to measure gap; flexible endoscopy or MRI to define displacement. Bladder neck appearance does not predict function.
- Endoscopic "cut-to-the-light" fails for gaps ≥1 cm and creates false passages; repeated urethrotomy complicates repair.
- **Perineal excision and primary anastomosis** is treatment of choice. Manoeuvres in order: corporal splitting → inferior pubectomy → supracrural rerouting. Abdominoperineal approach for fistula, bladder neck injury, failed repair, children. Total pubectomy for fistula or marked displacement. Limit lithotomy ≤5 h.

Complications

- ED in up to 82% — mainly from the **injury** (neurogenic, arterial), not treatment. Predictors: symphyseal diastasis, lateral urethral displacement, long gap. Arterial insufficiency may predispose to restenosis via bulbar ischaemia — consider penile duplex Doppler in at-risk men.
- Restenosis 5–15% — DVIU (direct visual internal urethrotomy) often works here. Incontinence <4% (bladder neck dependent). Follow for ED and incontinence ≥1 year.

8 · ANTERIOR URETHRAL INJURY

- Mostly **straddle injury** crushing the bulbar urethra against the pubis. Breach of Buck's fascia → butterfly perineal/scrotal haematoma. Main sequel: stricture, possibly years later.
- Classification: contusion, incomplete, complete disruption. Contusion/incomplete → urethral catheter. Major straddle injury → suprapubic tube.
- Low-velocity gunshot → primary repair; high-velocity → suprapubic diversion, delayed repair. Spongiosum has robust blood supply — debride minimally.
- Delayed: RUG + voiding cystourethrogram ± urethral ultrasound to measure scar. **Excision and primary anastomosis** for the obliterated 1.5–2 cm bulbar segment (>95% success); urethrotomy through obliteration is futile.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Multi-institutional PFUI study	Prospective, suprapubic tube vs early endoscopic realignment	No difference in stenosis or urethroplasty need	Suprapubic tube remains standard; realignment optional, brief

10 · MCQ ONE-LINERS

1. Tunica albuginea thinnest ventrolaterally.
2. Penile fracture deviates away from the tear.
3. Eggplant deformity: Buck's fascia intact.
4. Penile fracture urethral injury: ~20% West, ~3% Middle East.
5. Urethral check at fracture repair: flexible cystoscopy.
6. Penile fracture repair delay up to 7 days: outcome unchanged.
7. Penile gunshot: urethral injury 15–50%.
8. Human bites: leave open; dog bites: close primarily.
9. Penile replantation: dorsal artery with 11-0 nylon.
0. Amputation: double-bag; psychiatric referral always.
1. Testicular salvage >90% if explored within 72 hours.
2. Haematocele: up to 80% hide rupture.
3. Dislocated testis: superficial inguinal pouch.
14. Penile STSG non-meshed; scrotal STSG meshed.
15. Bladder injury: pelvic fracture in 83–95%.
16. Pelvic fracture: bladder injury in 5–10%.
17. Cystography: gross haematuria + pelvic fracture.
18. Cystography fill volume: 350 mL.
19. Plain cystography needs drainage film; CT cystography does not.
20. Extraperitoneal extravasation: flame-shaped.
21. Extraperitoneal rupture: 22-Fr catheter, cystogram at 2–3 weeks.
22. Intraperitoneal rupture: immediate repair.
23. PFUI disrupts the bulbomembranous junction.
24. PFUI acute: suprapubic tube.
25. PFUI urethroplasty at 2–3 months.
26. PFUI gap is stenosis, not stricture.
27. Straddle injury: bulbar urethra; anastomotic urethroplasty.

62 Paediatric Genitourinary Trauma

Reading time ≈ 37 min (slow pace)

1 · RENAL INJURY: HOW THE CHILD DIFFERS

- Kidney = commonest site of paediatric genitourinary (GU) trauma. Overwhelmingly **blunt** (road traffic, falls, bicycles, sport) and mostly grade 1–3; half have other organ injuries.
- Nephrectomy (3–5%) is **more likely in adult than paediatric trauma units**.
- Congenital anomalies (hydronephrosis, ureteropelvic junction [UPJ] obstruction) are found 3–5 × more often than in adults on trauma computed tomography (CT).

Feature	Child	Adult
Renal protection	Less perirenal fat, weaker abdominal wall, kidney below rib cage	Better cushioned
Blood pressure after haemorrhage	Maintained until late (≈ **25–30% volume loss**) — hypotension is a late sign	Falls earlier; reliable
Haematuria	May be absent with significant injury (up to 2/3 of grade ≥ 2 have normal urinalysis)	Classic imaging trigger
Imaging decision	**Mechanism-led**	Haematuria- and hypotension-led
Radiation	Cancer risk greater the younger the child → ALARA (as low as reasonably achievable)	Less weight

2 · INDICATIONS FOR IMAGING

Setting	Indication (American Urological Association [AUA] Urotrauma guideline)
Stable blunt	Gross haematuria
Stable blunt	Microscopic haematuria + systolic blood pressure (SBP) < 90 mm Hg
Mechanism / examination	Rapid deceleration; significant flank blow; rib fracture; flank ecchymosis
Penetrating	Any injury to abdomen, flank or lower chest — even without haematuria

- Paediatric mechanism triggers: fall > **10 feet (3 m)**, high-speed motor vehicle collision, blow to abdomen or flank. Associated predictors: thoracic or intra-abdominal organ injury; rib, spine, femur or pelvic fracture.
- Haematuria + hypotension + associated injury + mechanism together give > **98% sensitivity** for significant GU injury in children. European Association of Urology (EAU) agrees, adding associated abdominal injury.
- † Adult SBP < 90 mm Hg does not fit young children; Paediatric Advanced Life Support defines hypotension at 1–10 years as SBP < 70 + (2 × age in years) mm Hg.

> **KEY POINT** Trap: "no haematuria, so no imaging". Haematuria is only one of four routes into imaging; in children the two adult markers fail precisely, so mechanism must drive the decision.

3 · IMAGING STUDIES

Modality	Strength	Weakness / role
Focused assessment with sonography for trauma (FAST)	Free fluid; normal FAST + 24 h serial examination excludes significant injury	Does not examine the kidney (sensitivity 22–96%); not a staging test
Dedicated renal ultrasound (US) ± colour Doppler	No radiation; detects grade 3–5 reliably	Misses insignificant grade 1. AUA: may be used initially; EAU Paediatric: screening

Modality	Strength	Weakness / role
Intravenous (IV) contrast CT, immediate + delayed phases	Most sensitive and specific in stable patient	**Preferred**; radiation

- Children often get a **single phase** to limit radiation → misses urinary extravasation and ureteral injury. Solution: radiologist reviews the nephrographic phase in real time and adds a **delayed (excretory) phase** only if a leak is suspected (perinephric fluid, deep or multiple lacerations). The delayed phase exists to find urine, not blood.

4 · GRADING

American Association for the Surgery of Trauma (AAST) Organ Injury Scale (OIS): grade predicts intervention in adults and children. Grades 1–3 = **low grade** (intervention and functional loss negligible); 4–5 = **high grade**.

Grade	1989 AAST	2011 revision
1	Contusion or non-expanding subcapsular haematoma	Subcapsular haematoma/contusion
2	Perirenal haematoma; cortical laceration < 1 cm	Same, within Gerota fascia
3	Laceration > 1 cm, no collecting-system injury	Same, into medulla
4	Laceration into collecting system; main artery/vein injury with contained haemorrhage	Collecting system with extravasation; segmental vessel injury; pelvis laceration or **UPJ disruption**; shattered kidney
5	**Shattered kidney**; hilar avulsion	Main artery/vein laceration, avulsion or thrombosis

> **KEY POINT** Grades 1–3 differ only in depth (1 cm) and never involve the collecting system — entry into it makes grade 4. Shattered kidney is grade 5 (1989) but grade 4 (2011).

High-risk CT criteria within grade 4–5

Criterion	Predicts
Perirenal haematoma > 3.5 cm adult, > **2.2 cm child**	Bleeding needing angioembolisation
Vascular contrast extravasation; medial laceration	Bleeding
> 25% parenchyma devitalised	Bleeding and open surgery
Medial urinary extravasation	UPJ disruption
Extravasation + no contrast in distal ureter	Urine leak needing intervention

Bleeding risk is additive: 0–1 factor → < 10% intervention; 2–3 → > 50% (grade 4a vs 4b).

5 · NON-OPERATIVE MANAGEMENT (NOM)

- **Default at every grade**, including grade 4–5 and stable penetrating injury (strong recommendation, EAU Paediatric and Pediatric Trauma Society). Nephrectomy rates are higher with exploration than with NOM.
- Eligibility: stable, or stable after resuscitation.

ALGORITHM
Confirm stability → grade injury → bed rest → serial haemoglobin/haematocrit + abdominal examination → mobilise when gross haematuria resolves and counts are stable.

- Intensity scales with severity: grade 1–2, no gross haematuria → **23-hour observation**; high-risk grade 4 → intensive care, bloods 4–6-hourly, embolisation ready.

- **Early repeat imaging**: AUA — CT at 48 h for grade 4–5; EAU Paediatric — US at 2–3 days. In children it did not change outcomes; **symptoms, not imaging, predict complications**.
- Complications: grade 1–3 without devitalised segment ≈ zero. Grade 3 (devitalised) to 5: ≈ **25%** bleeding needing embolisation, ≈ **15%** urinoma needing intervention, ≈ 5% open exploration.

6 · COMPLICATIONS DURING NOM

Persistent urine leak / urinoma

- Only grade 4–5 leak; usual source is the parenchymal laceration (also pelvis, ureter, UPJ). Up to **85%** resolve spontaneously → managed non-operatively first (AUA, EAU).
- Symptomatic urinoma triad: ipsilateral **flank pain, ileus, low-grade fever**. Large symptomatic → early drainage; small → observe.
- **No contrast in the distal ureter** → cystoscopy + retrograde pyelography (RPG) to exclude UPJ disruption or major pelvic tear; many stent even if RPG is normal, as this finding predicts symptomatic urinoma.

Intervention (least invasive first)	Pros	Cons
Ureteral stent + Foley (AUA first line; ≈ 90%)	No external tube	Anaesthesia twice; stent symptoms
Percutaneous drain (large urinoma, failed stent)	Rapid relief; removed without anaesthesia	Dislodgement
Nephrostomy ± antegrade stent	When lithotomy impossible	Renal injury, bleeding
Embolise isolated segment	Segment not continuous with collecting system	Very rare
Open repair: debride → close collecting system (absorbable) → parenchyma over bolster → stent + drain	Definitive	Risk of nephrectomy

Stent removed at 4–6 weeks.

Persistent or delayed bleeding

	Persistent	Delayed
Mechanism	Vessels fail to tamponade	Arteriovenous fistula or pseudoaneurysm ruptures as haematoma liquefies
Timing	From injury	Typically **1–2 weeks** (up to 1 month)
Stable	Allow 2–3 transfusions, then embolise	Early embolisation unless mild
Unstable	Urgent angioembolisation	

- **Superselective angioembolisation** succeeds in > 80%; failure → repeat if stable → explore if still bleeding. Retained differential function: grade 4 ≈ 35–45%, grade 5 ≈ 10–30%.
- Post-embolisation syndrome (pain, fever, ileus) settles in 3–4 days; persistent fever → CT for abscess.

7 · OPERATIVE MANAGEMENT

- Indications: **haemodynamic instability from renal bleeding**; pulsatile or expanding retroperitoneal haematoma at laparotomy; bleeding uncontrolled by embolisation.
- Stable non-pulsatile haematoma at laparotomy → **leave alone**, even after gunshot wounds.
- **One-shot intraoperative intravenous pyelogram**: 2 mL/kg contrast → single film at 10–15 min. Confirms only that a **contralateral kidney** exists — informs nephrectomy vs reconstruction; does not grade.

- Early vascular control (incision medial to inferior mesenteric vein, over aorta): the one randomised trial showed no fall in nephrectomy; sensible with an expanding haematoma.

ALGORITHM
Contralateral kidney confirmed → open Gerota fascia → control main vessels → expose whole injury → debride → oversew vessels, close collecting system (absorbable) → close parenchyma over bolster/omentum (sliding-clip renorrhaphy) → drain. Penetrating: inspect ipsilateral ureter.

- **Nephrectomy**: irreparable injury or tiny remnant; or unstable (hypothermia, coagulopathy, ongoing loss) — speed saves life.
- **Renal artery laceration/avulsion**: end-artery organ, no collaterals → unstable → expect nephrectomy; repair only for solitary kidney or bilateral injury.
- **Renal artery occlusion** (intimal flap, embolism): stable → observe (post-infarction fever, pain; watch hypertension); endovascular stent an option; nephrectomy for uncontrolled pain or infection.

UPJ disruption

- Mechanism: deceleration + hyperextension — mobile kidney pulled from fixed ureter. Associated with **Chance fracture** (flexion–distraction), aortic and duodenal injury. † Chance fracture is the classic lap-belt injury.
- Haematuria may be absent; diagnosis often delayed (fever, ileus, urine in drains, abscess).
- CT: **medial extravasation, no distal ureteric contrast, often no laceration** → RPG.
- Stable and early → immediate spatulated tension-free ureteropelvic anastomosis over double-J stent (4–6 weeks) + drain. Unstable or diagnosed > 1 week → nephrostomy + drain, repair at 2–3 months.

Renal pelvis rupture and congenital anomalies

- Pelvis rupture: stent ± drain, or repair; with UPJ obstruction → delayed pyeloplasty.
- UPJ obstruction: injuries mostly grade 1–2, but leaks drain poorly → earlier stent ± drain. Then pyeloplasty 8–12 weeks later, or with stent in place (avoids infection after stent removal).
- Duplex kidney: an upper-pole ureteric stent will not drain a lower-pole leak.

8 · FOLLOW-UP AFTER RENAL INJURY

- Scarring: grade 1–2 none (no repeat imaging); grade 3 ≈ 50–60%; grade 4–5 ≈ 100%. Authors (ALARA): **US at 2–4 months** for grade 3–5; dimercaptosuccinic acid (DMSA) scan for suspected functional loss, hypertension or after embolisation.
- **Hypertension** (> 30 days after injury): rare in children; risk rises with grade and arterial injury. Mechanisms: ischaemia → renin; **Page kidney** (subcapsular collection compressing parenchyma); arteriovenous fistula; scarring. Treat with angiotensin-converting enzyme inhibitor or angiotensin receptor blocker (renin-mediated); nephrectomy for a poorly functioning scarred kidney.
- Yearly blood pressure after grade 3–5 or with solitary kidney; chronic kidney disease is rare.
- **Solitary kidney and sport**: organised sport allowed, padding for contact sport (American Academy of Pediatrics); counsel against motorised sport (all-terrain vehicles, dirt bikes) — higher risk.

9 · URETERAL TRAUMA

- Rare; mostly penetrating. **Haematuria absent in 25–50%**; ≈ 10% missed at laparotomy; missed injury → more nephrectomy.
- High-velocity gunshot: a contused-looking ureter may necrose days later → stent and drain if in doubt.
- Diagnosis: **delayed-phase CT** (complete transection = no distal contrast) or RPG.

AAST grade	Injury
I	Contusion/haematoma, no devascularisation

AAST grade	Injury
II / III	< 50% / ≥ **50%** transection
IV	Complete transection, < 2 cm devascularisation
V	Avulsion, > **2 cm** devascularisation

Immediate diagnosis (≤ 1 week)	Delayed (> 5–7 days)
Stable → repair under same anaesthetic	Try stent across injury (EAU: endourology first)
Unstable → damage control: ligate + nephrostomy, or drain	Fails → nephrostomy ± drain, reconstruct at 6–12 weeks

Level	Repair
Very proximal	Ureteropelvic anastomosis
Mid/upper	Spatulated ureteroureterostomy; Boari flap. **Avoid transureteroureterostomy in children** — long life risks the other kidney
Distal	Reimplantation ± psoas hitch
Long	Ileal ureter, Yang–Monti, buccal graft, autotransplant, nephrectomy

Follow-up: stent out → US at 4–6 weeks → yearly US for 2–3 years → hydronephrosis: mercaptoacetyltriglycine (MAG3) diuretic renogram → stricture: antegrade + retrograde study.

10 · BLADDER INJURY

Feature	Child	Adult
Bladder position	**More intraperitoneal** → isolated bladder injury commoner	Pelvic organ
Pelvic fracture	Half as frequent (pliable pelvis) → implies higher energy → more urethral injury	Commoner
Bladder injury without fracture	≈ half	Less common
Posterior urethral injury level	Supraprostatic/prostatic; **bladder neck injury commoner**	Bulbomembranous junction

- Triggers: **gross haematuria or retention**; pelvic fracture + gross haematuria → ≈ half have rupture.
- Delayed-phase CT may miss rupture → **retrograde cystography** (CT or plain), with retrograde urethrography (RUG).
- AAST: I contusion; II extraperitoneal < 2 cm; III extraperitoneal ≥ 2 cm or intraperitoneal < 2 cm; IV intraperitoneal ≥ 2 cm; V into bladder neck or trigone.

	Extraperitoneal	Intraperitoneal
Default	Catheter drainage	**Repair** at laparotomy (uncontained urine)
Repair if	Bone fragments in bladder, pelvic fixation hardware (infection), rectal/vaginal tear, other abdominal surgery, bladder neck	Always
Catheter	1–2 weeks → cystogram at removal	

> **KEY POINT** Bladder neck avulsion is commoner in children; girls depend more on the bladder neck for continence, so outcome is poor — many need secondary procedures, even closure and diversion.

11 · URETHRAL INJURY

Grade/type	AAST	Goldman
I	Contusion: meatal blood, normal RUG	Posterior urethra stretched, intact
II	Stretch, no extravasation	Rupture above intact urogenital diaphragm
III	Partial: extravasation, contrast reaches bladder	Rupture through diaphragm into bulbar urethra
IV	Complete, no bladder filling, < 2 cm gap	Bladder rupture extending into proximal urethra
V	≥ 2 cm gap or into prostate/vagina	Anterior urethral injury

- Pelvic fracture → posterior injury; straddle → bulbar. Retention, blood at meatus or vagina → **RUG before any catheter attempt** (AUA).
- Catheter passed → probably partial → confirm, drain, reassess; resultant stricture recurs in half after urethrotomy — no repeated endoscopy. Not passable → **suprapubic tube**.
- Endoscopic realignment: brief attempt only (irrigation enlarges haematoma). Immediate urethroplasty: EAU advises against (releases tamponade, over-debridement → impotence, incontinence). **Bladder neck injury is repaired at once.**

ALGORITHM
Suprapubic diversion → wait several months (haematoma resolves, bladder descends, gap shrinks) → cystogram + RUG ± endoscopy via both routes → transperineal excision and primary anastomosis → tension: partial pubectomy, bladder mobilisation → catheter 2–3 weeks → voiding trial, cystogram → suprapubic tube out.

- Children usually achieve tension-free perineal anastomosis; defects < 5 cm suit primary anastomosis.
- Complications as adults but lower: recurrent stricture ≈ 10% (urethrotomy if < 1 cm, else redo), incontinence ≤ 20% (proximal/bladder-neck injury), erectile dysfunction up to half (gap > 2.5 cm, lateral prostatic displacement, acute open realignment). Follow ≥ 1 year.
- **Girls**: rare (≈ 5% of female pelvic fractures), usually complete below bladder neck, with vaginal or rectal injury → suprapubic drainage, delayed primary anastomosis; continent if bladder neck spared.

12 · GENITAL INJURY

KEY POINT Most are accidental, but **abuse must always be assessed** (≈ 1 in 7 boys, 1 in 3 girls abused). Examine anus and rectum; involve social services before discharge.

Penis

- AAST: I skin; II Buck fascia, no tissue loss; III avulsion, glans/meatus laceration, defect < 2 cm; IV partial penectomy or defect ≥ 2 cm; V total penectomy.
- **Circumcision** (commonest cause): severe injury with untrained operators and the Mogen clamp (no glans shield); **careful lysis of preputial adhesions** prevents most. Excess skin removal → heals by secondary intention → *Paediatric Urology book, Ch 55*.
- **Hair/thread tourniquet**: slow, often painless, hair invisible → urethral fistula or amputation. Proximal-shaft fistula → catheter then multilayer closure; circumcision fistula is subcoronal → repair as hypospadias.
- Amputation: reattach (± microvascular); loss → neophalloplasty.
- Burns: full-thickness unmeshed shaft graft from sites hairless after puberty. Zipper: mineral oil or cut teeth. Bites: irrigate, secondary healing.

Scrotum and testis

- Testis AAST: I contusion; II subclinical tunica albuginea tear; III tear, < 50% loss; IV ≥ 50% loss; V destruction/avulsion.

- US signs of rupture: tunica albuginea discontinuity, haematocele, focal loss of Doppler flow → **explore, debride, close tunica albuginea** (AUA, EAU).
- Penetrating, tunica vaginalis intact → washout and closure in emergency department; breached → theatre. US at 6 weeks (exclude masked tumour).

Vagina and vulva

- **Straddle** ≈ 80%: labial, external, heals well, rarely theatre. Injury to hymen, vagina, perianal region or adjacent organs implies **penetration → abuse evaluation**.
- Examination under anaesthesia (cystoscopy, vaginoscopy, anoscopy) for heavy bleeding, poor cooperation or suspected abuse.
- Hymenal and superficial injuries heal rapidly without scar → **a normal examination does not exclude abuse**.

13 · MCQ ONE-LINERS

1. Commonest organ in paediatric GU trauma: kidney.
2. Children maintain blood pressure until ≈ 25–30% blood loss.
3. Up to two-thirds of grade ≥ 2 injuries have normal urinalysis.
4. Paediatric imaging is mechanism-led: fall > 10 feet, high-speed collision.
5. Penetrating flank injury needs imaging even without haematuria.
6. Delayed CT phase detects urinary extravasation.
7. Paediatric high-risk perirenal haematoma: > 2.2 cm.
8. Non-operative management is the default at every renal grade.
9. Up to 85% of urine leaks resolve spontaneously.
0. Urinoma triad: flank pain, ileus, low-grade fever.
1. First-line for symptomatic urinoma: ureteral stent.
2. Delayed renal bleeding: usually 1–2 weeks, pseudoaneurysm or fistula.
13. One-shot pyelogram: 2 mL/kg, film at 10–15 minutes.
14. Stable non-pulsatile haematoma at laparotomy: do not explore.
15. UPJ disruption: medial extravasation, empty distal ureter.
16. Post-traumatic hypertension is renin-mediated: block the renin–angiotensin system.
17. Ureteral injury: haematuria absent in 25–50%.
18. Child's bladder more intraperitoneal: isolated bladder injury commoner.
19. Intraperitoneal bladder rupture: always repair.
20. Paediatric posterior urethral injury: often prostatic or bladder neck.
21. Pelvic fracture urethral injury: suprapubic tube, delayed perineal anastomosis.
22. Mogen clamp: highest risk of glans amputation.
23. Normal genital examination does not exclude abuse.

★ Putting it together: stones, obstruction and trauma decisions

Reading time ≈ 10 min (slow pace) · Integrates Ch 54, 62, 93–100, 125 · use these as the skeleton of long-essay answers

A · RENAL COLIC IN THE EMERGENCY DEPARTMENT

ALGORITHM

1. Analgesia first: a non-steroidal anti-inflammatory drug (NSAID; ketorolac 30 mg) unless renal failure, bleeding or pregnancy. Non-contrast computed tomography (NCCT) confirms; ultrasound first in pregnancy and children (→ *Ch 96*).

2. Decompress now (retrograde stent or percutaneous nephrostomy, equally effective) if obstruction with infection, bilateral obstruction, solitary kidney, immunosuppression, impending renal deterioration, refractory pain or intractable vomiting. Treat the stone later — never definitive lithotripsy in an infected obstructed kidney.

3. Otherwise stone <10 mm → trial of passage ± medical expulsive therapy (α-blocker; benefit confined to distal stones 5–10 mm) for 4–6 weeks → if it fails, ureteroscopy (URS) or shock wave lithotripsy (SWL) (→ *Ch 97*).

B · CHOOSING THE STONE OPERATION

Stone	First choice	Notes
Ureteric, any level	URS or SWL	URS: higher stone-free rate in one session; SWL: more retreatment, fewer complications. Large hard proximal stones >15–20 mm → consider percutaneous nephrolithotomy (PCNL) ± antegrade URS
Renal <2 cm, not lower pole	SWL or URS	SWL works best with density <900 Hounsfield units and skin-to-stone distance <10 cm
Lower pole >1 cm	PCNL or URS	SWL not first line — fragments clear poorly (Lower Pole I trial)
Total burden >2 cm, staghorn	PCNL	Staged URS an option; SWL not first line

KEY POINT SWL absolute contraindications: pregnancy, uncorrected coagulopathy or antithrombotic drugs, untreated urinary tract infection, uncontrolled hypertension, aortic or renal artery aneurysm, and obstruction distal to the stone. Do not stent routinely after SWL or uncomplicated URS (→ *Ch 98*).

C · PREVENTING THE NEXT STONE

ALGORITHM

Everyone: fluids for urine output ≥2.5 L/day †, normal calcium intake (restriction increases stones), low sodium, less non-dairy animal protein, more fruit and vegetables (→ *Ch 99*).

High risk or recurrent → stone analysis, serum chemistry and one or two 24-hour urines → targeted therapy → repeat 24-hour urine within 6 months.

Abnormality	Treatment
Hypercalciuria	Thiazide + potassium citrate (thiazide-induced hypokalaemia lowers citrate)
Hypocitraturia	Potassium citrate
Uric acid stones	Alkalinise to pH 6.5–7 †; allopurinol only with hyperuricosuria/gout or failed alkalinisation
Cystine	≥3 L/day, alkalinise to pH 7.0–7.5, then thiol (tiopronin)

Abnormality	Treatment
Struvite	Complete stone clearance; most also have a metabolic abnormality

D · RENAL TRAUMA

ALGORITHM

Image (contrast CT with delayed phase) for gross haematuria, microhaematuria with shock, deceleration injury or penetrating flank/abdominal trauma; blunt microhaematuria without shock → observe (→ *Ch 94*).

Stable, staged grade I–IV and selected V → non-operative management (>90% success in blunt injury). Contrast blush → angioembolisation. Symptomatic urine leak → retrograde stent + urethral catheter.

Explore for instability despite resuscitation or an expanding/pulsatile haematoma. Unstable at laparotomy without CT → one-shot intravenous pyelogram (2 mL/kg, film at 10 min) to confirm a working contralateral kidney.

E · BLADDER AND URETHRA

Injury	Management
Extraperitoneal bladder rupture (pelvic fracture)	22 Fr urethral catheter; cystogram at 2–3 weeks. Repair if at laparotomy anyway, bone spicule, or not healed by 4 weeks
Intraperitoneal bladder rupture (blow to a full bladder)	† Formal surgical repair
Pelvic fracture urethral injury	Suprapubic tube now (or brief primary realignment if stable) → delayed urethroplasty
Penile fracture	Prompt exploration and tunical repair; look for urethral injury (→ *Ch 125*)
Testicular rupture	Immediate exploration and repair

Cystography must be retrograde, filled to at least 350 mL; clamping the catheter during intravenous-contrast CT is not enough.

F · URETERIC DEFECT: WHICH REPAIR?

Defect	Repair
2–3 cm, upper/mid	Ureteroureterostomy (spatulated, tension-free)
Lower ureter, 6–10 cm	Psoas hitch reimplant
10–15 cm	Boari flap
Longer or whole ureter	Transureteroureterostomy, ileal ureter (creatinine <2 mg/dL), autotransplantation, or nephrectomy (→ *Ch 93*)

G · CHILDREN ARE DIFFERENT

KEY POINT **Stones:** ultrasound first, low-dose NCCT only if ultrasound is negative and suspicion stays high; every child gets a full metabolic evaluation after the first stone (monogenic disease is common). Renal ≤20 mm → SWL or URS; >20 mm → PCNL (→ *Ch 54*). **Trauma:** imaging is driven by mechanism (rapid deceleration, fall >3 m, flank blow, rib or spine fracture), not by haematuria or hypotension, because children hold their blood pressure until late; a perirenal haematoma >2.2 cm predicts bleeding needing angioembolisation (→ *Ch 62*).